Supplementary Material

Systematic robustness of a boosted top-quark tagger on ATLAS open data: a constituent-level feature reconstruction, paired SHAP-Delta attribution drift, and conformal coverage

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S1 Scope and provenance

This document contains the complete result tables referred to in the main text. Every value is taken from the frozen publication artifacts of a single analysis chain, identified by

- reconstruction fingerprint `c28a262540824e9db90f34c421f10c70` (810 cache files, 80,481,421 jet entries, seventeen samples);
- final XGBoost model SHA-256 `63a23308efbf2ce90418c8e657730f57 1db4798fc9ca2fb3692cb986de405f4c`, developed from a five-million-jet nominal subset of which 4,499,612 jets were used for fitting and 500,385 were held out for validation and conformal calibration;
- metrics artifact `final_metrics_atlas_tagger_xgb_final_production_round1_round1.json`;
- paired SHAP-Delta campaign `shap_delta_final_production_round3`;
- conformal campaign `conformal_coverage_round1`;
- pairing logs `ATLAS80030_pair_all_round1` and `ATLAS80030_pair_bias_round1`;
- for the robustness extension of Tables S8–S10, the result folders `stat_bootstrap_round1`, `tagger_variants_round1` and `feature_response_round1`. The bootstrap and the seed study each first reproduce the frozen result exactly from the frozen model and predictions; the input-response study uses no model, and its ECF_1 response reproduces the stored-branch audit of Table S1.

Tables S2–S7 are produced from those artifacts by `tables/make_supplementary_tables.py`. Table S1 comes from the completed full-sample `03_Code_files/stale_hlf_audit.py` outputs. The regrouping script for Tables S2–S7 copies every number verbatim and asserts that the multiset of numeric tokens in its outputs is exactly the multiset in the frozen sources, so a value cannot be dropped, altered or introduced by the layout change. The paired SHAP-Delta outputs are additionally cross-checked value by value against the frozen twenty-column artifact of the previous revision.

Table S1 the stored high-level branch audit that supports Section 2.3 of the main text.

Tables S2 and S3 the bin-by-bin systematic breakdown underlying Fig. 2(b) of the main text, split into its detector part and its modelling and scale part.

Tables S4 and S5 the complete conformal result underlying Fig. 4 of the main text, split into coverage and prediction-set structure.

Tables S6 and S7 the complete paired SHAP-Delta result underlying Fig. 3 and Table 5 of the main text, split into per-feature attribution results and systematic-level score and rank summaries.

Table S8 the bin-by-bin statistical standard deviations behind the bands of Fig. 2 of the main text.

Table S9 every run of the training-seed and input-set study summarised in Tables 3, 4 and 7 of the main text.

Table S10 the full per-jet input response summarised in Table 6 of the main text.

Tables S8–S10 are written by `tables/make_extension_tables.py`, which refuses to write unless the bootstrap central values and the seed-42 control run reproduce the frozen numbers exactly and every summary statistic recomputes from the per-replica or per-run values.

S2 Stored high-level branches in the varied samples

For the four calorimeter-cluster variations with validated exact pairing, the fifteen stored `fjet_*` high-level branches are float32 bit-identical to nominal for every matched jet, although the released constituents change. Table S1 reports 5,000,000 exact matches for each of `esup`, `esdown` and `cpos`, and 4,995,059 for `cer`. Across all eligible matched-jet/branch comparisons, the exact identity counts are 75,000,000/75,000,000 for each fully matched sample and 74,925,885/74,925,885 for `cer`.

The relative summed-constituent-p_{T} shift has mean +6.780753% and median +7.406764% under `esup`; the corresponding `esdown` mean and median are −6.780749% and −7.406822%. The `cer` summed-energy response has RMS 3.770123%, and the median `cpos` constituent-centroid displacement is $\Delta R = 4.9387 \times 10^{-5}$. The track samples and `bias` are not pairable on the full sample: their exact-key-hit counts are diagnostic only. Track constituent multiplicity is therefore compared as an unpaired sample distribution; no pairwise stored-branch identity or same-jet constituent-change fraction is inferred for those five samples.

Table S1: Full-sample audit of stored high-level branches in the nine detector variations. The four calorimeter-cluster samples pass the exact-key pairing preflight; for every valid matched jet, all fifteen stored high-level values are identical to nominal as float32 bit patterns. Exact-key hits for track samples and `bias` are diagnostics only and do not support pairwise identity fractions.

Sample	Shards	Exact-key hits	Pairing basis	HLF identity	Constituent evidence
`esup`	50	5,000,000	exact key (100.000%)	75,000,000/75,000,000 (100.000%)	median $\Delta\Sigma p_{\mathrm{T}}$: +7.4068%
`esdown`	50	5,000,000	exact key (100.000%)	75,000,000/75,000,000 (100.000%)	median $\Delta\Sigma p_{\mathrm{T}}$: -7.4068%
`cer`	50	4,995,059	exact key (99.901%)	74,925,885/74,925,885 (100.000%)	RMS $\Delta\Sigma E$: 3.7701%
`cpos`	50	5,000,000	exact key (100.000%)	75,000,000/75,000,000 (100.000%)	median centroid ΔR: 4.9387×10^{-5}
`teg`	50	2,250,423	N/A / not pairable	N/A	median N_{c}: 49 vs nominal 50 (unpaired)
`tej`	50	1,906,619	N/A / not pairable	N/A	median N_{c}: 49 vs nominal 50 (unpaired)
`tfl`	50	2,749,571	N/A / not pairable	N/A	median N_{c}: 48 vs nominal 50 (unpaired)
`tfj`	50	2,006,933	N/A / not pairable	N/A	median N_{c}: 49 vs nominal 50 (unpaired)
`bias`	50	4,167,067	N/A / not pairable	N/A	N/A

Definitions: $\Delta\Sigma p_{\mathrm{T}} = \Sigma p_{\mathrm{T}}^{\mathrm{sys}} / \Sigma p_{\mathrm{T}}^{\mathrm{nom}} - 1$; $\Delta\Sigma E = \Sigma E^{\mathrm{sys}} / \Sigma E^{\mathrm{nom}} - 1$. The `cer` value is the RMS of the signed summed-energy response, not its near-zero signed median. The `cpos` value is the median ΔR of the constituent-p_{T}-weighted η–ϕ centroids. Track multiplicities are separate full-sample distributions, not matched-jet changes; the nominal median is 50. `teg` has 183,802 duplicate keys, and the other track samples and `bias` fall below the validated pairing-coverage floor. The full counts and per-branch comparisons are preserved in `04_Results/stale_hlf_audit/`.

S3 Bin-by-bin systematic uncertainties

The fourteen physical p_{T} bins span 350–3150 GeV in 200 GeV steps. All uncertainty columns are relative uncertainties on background rejection at 50% signal efficiency, in percent. Table S2 carries the cluster and track categories, Table S3 the generator modelling and ISR/FSR scale categories together with the total of all four. The statistical standard deviation of every entry is given in Table S8.

Table S2: Detector systematic uncertainties on background rejection at 50% signal efficiency, in each of the fourteen physical 200 GeV p_{T} bins. All uncertainty columns are relative uncertainties in percent. The nominal background rejection is repeated here so that each relative uncertainty can be read against the value it applies to.

p_{T} low	p_{T} high	Nominal BR	CES	CER	CPOS	Cluster total	Track eff.	Fake rate	Bias	Track total
350	550	93.414	9.97923	4.11817	1.54679	10.9058	3.78563	2.07996	1.73344	4.65425
550	750	66.9929	11.3853	2.31711	0.489907	11.629	3.15357	4.43777	0.608945	5.4781
750	950	58.5127	14.3254	3.8968	1.48534	14.9201	4.0226	2.93045	3.05909	5.84182
950	1150	53.7758	15.5448	1.04451	0.0182623	15.5799	1.72603	2.74085	0.92674	3.36901
1150	1350	55.0452	13.8141	3.44808	1.34744	14.3016	1.42879	4.61806	2.5554	5.4679
1350	1550	48.4567	13.3623	2.79696	0.909349	13.6822	2.99679	4.45887	1.76832	5.6559
1550	1750	49.1348	13.2405	3.47274	0.837172	13.7139	0.915461	2.82229	0.175584	2.97224
1750	1950	44.1886	14.2629	4.77981	1.38144	15.1058	1.54494	2.68994	0.422248	3.13064
1950	2150	40.8557	9.92577	1.8989	0.812175	10.1384	0.921182	4.53883	1.96547	5.03116
2150	2350	39.6283	12.6348	3.44452	1.26718	13.1571	4.11698	6.04944	2.89518	7.8694
2350	2550	37.5143	12.5682	2.31997	0.801577	12.8056	1.73761	5.46226	0.90764	5.80339
2550	2750	33.712	6.65472	1.2474	0.558901	6.79365	0.910098	4.01797	0.113344	4.12131
2750	2950	32.3081	8.18673	4.0166	1.71849	9.27948	2.19987	4.20073	2.37679	5.30421
2950	3150	28.5641	7.27987	0.0548925	0.259853	7.28472	3.14807	4.78966	0.284346	5.73864

p_{T} bin edges in GeV. CES is the `esup`/`esdown` envelope, track efficiency the `teg`/`tej` envelope and fake rate the `tfl`/`tfj` envelope; category totals are quadrature sums of the components to their left.

Table S3: Generator-modelling and ISR/FSR scale uncertainties on background rejection at 50% signal efficiency, in the same fourteen bins, with the total of all four categories. All columns are relative uncertainties in percent.

p_T low	p_T high	Signal model	Bkg model	Model total	Signal ISR	Signal FSR	Bkg ISR	Bkg FSR	ISR/FSR total	Total
350	550	20.2801	20.2063	28.6283	1.29504	20.399	7.22165	10.5631	24.1149	39.2645
550	750	14.1618	9.56575	17.0898	0.774375	18.349	6.88518	7.21155	20.8973	29.8999
750	950	8.39876	10.374	13.3476	0.528659	15.4347	5.0622	8.73264	18.4497	27.844
950	1150	9.35958	6.08504	11.1638	0.352858	16.4144	0.518628	5.87127	17.4441	26.1344
1150	1350	10.7978	1.46415	10.8966	0.440081	17.2831	3.44701	5.70108	18.5279	26.3904
1350	1550	10.7978	7.20897	12.9831	0.289097	17.3928	1.57054	5.95989	18.4548	26.9876
1550	1750	20.2801	7.25911	21.5401	0.244223	11.1576	1.89294	4.79582	12.2937	28.4959
1750	1950	20.2801	5.62898	21.0468	0.21374	10.6371	1.91893	3.8948	11.4911	28.5132
1950	2150	20.2801	4.21909	20.7143	0.367505	16.1712	0.924417	1.12223	16.2406	28.6521
2150	2350	20.2801	4.25508	20.7217	0.149738	14.252	1.18127	1.59226	14.3901	29.5212
2350	2550	20.2801	4.69118	20.8156	0.145985	8.02331	0.245851	1.03581	8.09494	26.3909
2550	2750	20.2801	7.14007	21.5003	0.0935954	10.6968	2.03069	0.639324	10.907	25.3844
2750	2950	20.2801	6.17343	21.1989	0.0594766	11.9783	2.16768	0.957883	12.2107	26.6972
2950	3150	20.2801	6.45282	21.282	0.0653595	5.29897	1.107	0.457086	5.43302	23.842

p_T bin edges in GeV. Signal modelling is measured only in the bins below 1.5 TeV and held at its maximum measured value from 1.55 TeV upwards; the 1350–1550 GeV interval carries the official plotting convention described in Section 4.1 of the main text and is not an independent measurement of that interval. The Total column is the quadrature sum of the cluster and track totals of Table S2 with the model and ISR/FSR totals here.

S4 Class-conditional split-conformal coverage

Thresholds were calibrated once, at target coverage 0.9, on the frozen held-out nominal validation split (500,385 jets; 252,215 signal, 248,170 background), giving $q_{\text{sig}} = 0.5700125098228455$ and $q_{\text{bkg}} = 0.6447482109069824$ on the nonconformity score $s_c = 1 - \hat{p}_c$. Class c enters a jet's prediction set when $1 - \hat{p}_c \leq q_c$, equivalently $\hat{p}_c \geq 1 - q_c$. The thresholds were applied unchanged to every sample below. Coverage is empirical and unweighted; no refitting, recalibration or reweighting was performed, and the calibration jets took no part in fitting the model or in selecting its hyperparameters.

The two class-conditional coverages are the primary quantities. Overall coverage is a mixture of them weighted by each sample's class composition, which ranges from 50.5% signal on `test_nominal` to 18.5% on `tfl`, and is reported as a descriptive number only. Deviations are quoted both from the 0.9 target and from the value measured on the nominal test sample.

Table S4: Class-conditional split-conformal coverage at target 0.9. The two class-conditional coverages are the primary quantities; the overall coverage is a descriptive number whose value depends on the class mixture of the sample, which differs strongly between the calorimeter and the track samples.

Dataset	n	n_s	n_b	Signal cov.	Bkg cov.	Overall cov.	Δ_s target	Δ_b target	Δ target	Δ_s nominal	Δ_b nominal	Δ nominal
test_nominal	10306812	5207478	5099334	0.900423	0.8999877	0.9002076	0.0004230071	-1.227611e-05	0.0002076491	0	0	0
esup	4999999	2592575	2407424	0.9011597	0.8960561	0.8987024	0.001159658	-0.003943884	-0.00129762	0.0007366508	-0.003931607	-0.001505269
esdown	4999999	2592653	2407346	0.8903914	0.9021154	0.8960362	-0.009608575	0.002115442	-0.003963821	-0.01003158	0.002127718	-0.00417147
cer	4999756	2590441	2409315	0.8989906	0.8994121	0.8991937	-0.001009442	-0.0005879264	-0.0008063193	-0.001432449	-0.0005756503	-0.001013968
cpos	4999999	2592597	2407402	0.9002356	0.8994331	0.8998492	0.0002355553	-0.0005669182	-0.00015082	-0.0001874518	-0.0005546421	-0.0003584691
teg	4999998	1500292	3499706	0.8994109	0.8981366	0.898519	-0.0005890853	-0.001863414	-0.001481041	-0.001012092	-0.001851138	-0.00168869
tej	4999999	1589974	3410025	0.9004927	0.8999667	0.900134	0.0004927125	-3.328421e-05	0.00013398	6.970539e-05	-2.10081e-05	-7.366905e-05
tfl	4999996	923044	4076952	0.898085	0.8999292	0.8995887	-0.001914968	-7.083723e-05	-0.0004112803	-0.002337975	-5.856112e-05	-0.0006189294
tfj	4999983	1599126	3400857	0.8957743	0.9003584	0.8988923	-0.004225683	0.0003583509	-0.001107744	-0.00464869	0.000370627	-0.001315393
bias	4999999	2597808	2402191	0.9005931	0.8997295	0.9001782	0.0005931154	-0.0002705447	0.00017818	0.0001701084	-0.0002582686	-2.946904e-05

Deviations are quoted as fractions, from the 0.9 target and from the value measured on `test_nominal`. Thresholds were calibrated once on the frozen nominal validation split and applied unchanged to every row.

Table S5: Structure of the class-conditional conformal prediction sets on the same samples: mean set size, and the rate and count of one-class, two-class and empty sets.

Dataset	Mean set size	Singleton rate	Ambiguous rate	Empty rate	Singleton n	Ambiguous n	Empty n
test_nominal	1.103904	0.8960961	0.1039039	0	9235894	1070918	0
esup	1.106734	0.893266	0.106734	0	4466329	533670	0
esdown	1.106356	0.893644	0.106356	0	4468219	531780	0
cer	1.104816	0.8951841	0.1048159	0	4475702	524054	0
cpos	1.104307	0.895693	0.104307	0	4478464	521535	0
teg	1.100242	0.8997582	0.1002418	0	4498789	501209	0
tej	1.099836	0.9001636	0.09983642	0	4500817	499182	0
tfl	1.097283	0.9027171	0.09728288	0	4513582	486414	0
tfj	1.100723	0.8992775	0.1007225	0	4496372	503611	0
bias	1.104136	0.895864	0.104136	0	4479319	520680	0

"Ambiguous" denotes a prediction set containing both classes. No sample produces an empty set.

S5 Complete paired SHAP-Delta results

SHAP values are exact TreeSHAP attributions of the frozen model in the additive margin space of the booster; score values are in probability space. "Mean |ΔSHAP|" is the mean over jets of the absolute per-jet change in a feature's attribution, and "relative attribution drift" is the relative change in that feature's global mean |SHAP|. These are different quantities and are reported separately.

Table S6 gives the per-feature results, which are the quantities that differ from feature to feature. Table S7, which follows it, gives the score-level and rank-level summaries, which are properties of a whole systematic, sample-size and class-subset group; in the previous revision these were repeated on all fifteen feature rows of each group, and separating them removes that repetition.

Table S6: Complete per-feature paired SHAP-Delta results, round 3. SHAP values are exact TreeSHAP attributions of the frozen model in the additive margin space of the booster. Mean |ΔSHAP| is the mean over jets of the absolute per-jet change in a feature's attribution; relative attribution drift is the relative change in that feature's global mean |SHAP|. The two are different quantities and are not interchangeable. Score-level and rank-level summaries, which are properties of a whole systematic–subset–sample-size group rather than of a feature, are given separately in Table S7.

Systematic	Size	Subset	Feature	n	Mean ΔSHAP	Median ΔSHAP	Mean |ΔSHAP|	Median |ΔSHAP|	Nom. mean |SHAP|	Sys. mean |SHAP|	Rel. attr. drift	Nom. rank	Sys. rank
esup	25000	all	τ_1^{WTA}	25000	-0.00434156	-0.00425814	0.0190665	0.012869	0.240339	0.235264	-0.0211173	11	11
esup	25000	all	τ_2^{WTA}	25000	-0.00731675	-0.00515284	0.0194375	0.0137851	0.28977	0.293205	0.0118541	6	6
esup	25000	all	τ_3^{WTA}	25000	-0.00367214	-0.0020184	0.0223208	0.0146312	0.263671	0.267294	0.0137407	10	9
esup	25000	all	τ_4^{WTA}	25000	0.0181638	0.0142685	0.0461951	0.0339434	0.915571	0.915845	0.000299139	1	1
esup	25000	all	$\sqrt{d_{12}}$	25000	0.0109114	0.00742533	0.0353833	0.0249318	0.264725	0.262701	-0.00764745	9	10
esup	25000	all	$\sqrt{d_{23}}$	25000	0.00694671	0.00527251	0.0418826	0.0294513	0.382672	0.377808	-0.0127097	5	5
esup	25000	all	ECF_1	25000	-0.0045179	-0.00805226	0.0513568	0.0284781	0.152907	0.152777	-0.000844817	12	12
esup	25000	all	ECF_2	25000	0.00284624	0.00170189	0.0153989	0.0095173	0.044636	0.0456228	0.0221074	15	15
esup	25000	all	ECF_3	25000	-0.013611	0.0101904	0.128456	0.0773123	0.712902	0.677895	-0.0491062	2	2
esup	25000	all	C_2	25000	0.000575124	6.85288e-05	0.0103149	0.00631687	0.0960636	0.0958936	-0.00176997	14	14
esup	25000	all	D_2	25000	-0.0114796	-0.00908495	0.0240718	0.0167221	0.285947	0.280028	-0.0207	7	7
esup	25000	all	L_2	25000	-0.00247824	-0.00235188	0.0472834	0.029814	0.411918	0.428734	0.0408259	3	3
esup	25000	all	L_3	25000	0.00307768	0.00332733	0.0204141	0.013195	0.139444	0.136245	-0.0229404	13	13
esup	25000	all	Q_w	25000	0.0177802	0.0116571	0.0402339	0.0263597	0.271452	0.272451	0.00368286	8	8
esup	25000	all	ThrustMaj	25000	-0.0112654	-0.00611538	0.0472179	0.0321531	0.409828	0.41075	0.00224971	4	4
esup	25000	signal	τ_1^{WTA}	12961	-0.00742859	-0.00598618	0.0180438	0.0129641	0.198022	0.190008	-0.0404674	10	11
esup	25000	signal	τ_2^{WTA}	12961	-0.00869531	-0.00659244	0.0192281	0.0141659	0.260284	0.26242	0.00820569	8	8
esup	25000	signal	τ_3^{WTA}	12961	-0.00495204	-0.00314064	0.0247753	0.0168893	0.292879	0.297476	0.0156964	6	7
esup	25000	signal	τ_4^{WTA}	12961	0.020185	0.0155597	0.0469984	0.0351849	0.877359	0.873818	-0.00403692	1	1
esup	25000	signal	$\sqrt{d_{12}}$	12961	0.00941356	0.00743981	0.0334983	0.0241757	0.197561	0.202804	0.0265387	11	9
esup	25000	signal	$\sqrt{d_{23}}$	12961	0.000580019	-0.000811756	0.0446346	0.0326138	0.351147	0.347798	-0.00953497	5	5
esup	25000	signal	ECF_1	12961	-0.0150534	-0.0115825	0.048538	0.0329386	0.146334	0.140155	-0.0422269	12	12
esup	25000	signal	ECF_2	12961	0.00137556	0.000689186	0.0138125	0.00890921	0.0342044	0.0359198	0.0501518	15	15
esup	25000	signal	ECF_3	12961	-0.054434	-0.0200152	0.164731	0.102863	0.628236	0.585732	-0.0676562	2	2
esup	25000	signal	C_2	12961	0.000765133	0.000294112	0.00950268	0.00608435	0.0804367	0.0800343	-0.00500249	14	13
esup	25000	signal	D_2	12961	-0.011811	-0.0107869	0.0230544	0.0171047	0.20712	0.193486	-0.0658275	9	10
esup	25000	signal	L_2	12961	0.00349563	-0.00143513	0.057718	0.0377151	0.407798	0.428608	0.0510304	4	4
esup	25000	signal	L_3	12961	0.00360274	0.00409439	0.015993	0.0109154	0.0821967	0.0772485	-0.0602001	13	14
esup	25000	signal	Q_w	12961	0.0184303	0.0123477	0.0489395	0.0353009	0.288984	0.297729	0.0302614	7	6
esup	25000	signal	ThrustMaj	12961	-0.0177852	-0.0114986	0.0565996	0.0416927	0.446352	0.441174	-0.0116006	3	3
esup	25000	background	τ_1^{WTA}	12039	-0.00101812	-0.00237626	0.0201676	0.0127597	0.285895	0.283984	-0.00668402	9	9
esup	25000	background	τ_2^{WTA}	12039	-0.00583262	-0.00374976	0.019663	0.0132174	0.321513	0.326348	0.0150382	8	8
esup	25000	background	τ_3^{WTA}	12039	-0.00229422	-0.00119336	0.0196784	0.0125813	0.232227	0.234798	0.0110714	11	11
esup	25000	background	τ_4^{WTA}	12039	0.0159877	0.0132152	0.0453303	0.0325543	0.956712	0.961086	0.00457163	1	1
esup	25000	background	$\sqrt{d_{12}}$	12039	0.012524	0.0074071	0.0374127	0.0258643	0.337036	0.327184	-0.0292321	7	7
esup	25000	background	$\sqrt{d_{23}}$	12039	0.013801	0.0101193	0.0389199	0.0259463	0.416606	0.410117	-0.0155771	3	4
esup	25000	background	ECF_1	12039	0.00682441	-0.00510362	0.0543914	0.0241138	0.159984	0.166366	0.0398924	13	13
esup	25000	background	ECF_2	12039	0.00442955	0.00289722	0.0171068	0.0103102	0.0558665	0.0560688	0.00362197	15	15
esup	25000	background	ECF_3	12039	0.0303384	0.0319794	0.0894031	0.0613401	0.804059	0.777115	-0.0335097	2	2
esup	25000	background	C_2	12039	0.000370563	-0.000122786	0.0111893	0.0066075	0.112887	0.112969	0.000721118	14	14

Systematic	Size	Subset	Feature	n	Mean ΔSHAP	Median ΔSHAP	Mean |ΔSHAP|	Median |ΔSHAP|	Nom. mean |SHAP|	Sys. mean |SHAP|	Rel. attr. drift	Nom. rank	Sys. rank
esup	25000	background	D_2	12039	-0.0111228	-0.00714368	0.0251672	0.0163062	0.37081	0.373197	0.00643682	5	6
esup	25000	background	L_2	12039	-0.00890962	-0.002859	0.0360496	0.0235629	0.416353	0.428868	0.0300589	4	3
esup	25000	background	L_3	12039	0.00251241	0.00220869	0.0251738	0.0168725	0.201075	0.19976	-0.00653881	12	12
esup	25000	background	Q_w	12039	0.0170803	0.0111133	0.0308616	0.0192962	0.252578	0.245239	-0.0290558	10	10
esup	25000	background	ThrustMaj	12039	-0.00424623	-0.0027098	0.0371177	0.0243137	0.370504	0.377996	0.0202232	6	5
esup	50000	all	τ_1^{WTA}	50000	-0.00435308	-0.00434025	0.0191103	0.0129801	0.239952	0.234746	-0.0216935	11	11
esup	50000	all	τ_2^{WTA}	50000	-0.00739281	-0.00518003	0.0193923	0.0136991	0.289399	0.29289	0.0120618	6	6
esup	50000	all	τ_3^{WTA}	50000	-0.00391045	-0.00221208	0.0224239	0.0148252	0.265045	0.26864	0.0135639	9	9
esup	50000	all	τ_4^{WTA}	50000	0.0178074	0.0143479	0.0462972	0.0338561	0.914396	0.915141	0.000815591	1	1
esup	50000	all	$\sqrt{d_{12}}$	50000	0.011045	0.00767586	0.0355069	0.0250531	0.263668	0.261946	-0.00653164	10	10
esup	50000	all	$\sqrt{d_{23}}$	50000	0.0069583	0.00545232	0.042014	0.02936	0.382003	0.377171	-0.0126507	5	5
esup	50000	all	ECF_1	50000	-0.00481076	-0.00800223	0.0514146	0.028577	0.153339	0.153231	-0.000702014	12	12
esup	50000	all	ECF_2	50000	0.00252231	0.00163595	0.0153134	0.00937944	0.0444037	0.0453558	0.0214414	15	15
esup	50000	all	ECF_3	50000	-0.0128796	0.00989418	0.127355	0.0762991	0.707832	0.673906	-0.0479295	2	2
esup	50000	all	C_2	50000	0.000523686	5.70007e-05	0.0103817	0.0063462	0.0960296	0.0958756	-0.00160371	14	14
esup	50000	all	D_2	50000	-0.0115232	-0.00902383	0.0240372	0.016671	0.286145	0.280275	-0.0205135	7	7
esup	50000	all	L_2	50000	-0.00247689	-0.00230951	0.0471149	0.0299817	0.412361	0.428791	0.0398445	3	3
esup	50000	all	L_3	50000	0.00322823	0.0033555	0.0202861	0.0131214	0.137862	0.134575	-0.023845	13	13
esup	50000	all	Q_w	50000	0.0176774	0.0118103	0.0402834	0.0266111	0.271573	0.272342	0.00283227	8	8
esup	50000	all	ThrustMaj	50000	-0.0117808	-0.00658655	0.0471487	0.0323781	0.410352	0.411292	0.00229136	4	4
esup	50000	signal	τ_1^{WTA}	25927	-0.00735453	-0.00609876	0.0181184	0.0131077	0.19842	0.190313	-0.0408576	10	11
esup	50000	signal	τ_2^{WTA}	25927	-0.00858766	-0.00646585	0.0190256	0.0140086	0.259692	0.261843	0.00828144	8	8
esup	50000	signal	τ_3^{WTA}	25927	-0.00498602	-0.00340959	0.0247696	0.0169639	0.294227	0.299049	0.0163885	6	6
esup	50000	signal	τ_4^{WTA}	25927	0.0201126	0.0157146	0.0470576	0.035239	0.875636	0.872878	-0.00314865	1	1
esup	50000	signal	$\sqrt{d_{12}}$	25927	0.00967909	0.00778377	0.0339121	0.0244445	0.197712	0.20323	0.0279083	11	9
esup	50000	signal	$\sqrt{d_{23}}$	25927	0.00106707	-0.000241786	0.0446774	0.0323065	0.351247	0.34773	-0.0100139	5	5
esup	50000	signal	ECF_1	25927	-0.0154546	-0.0118567	0.0486477	0.0330138	0.1458	0.139754	-0.0414668	12	12
esup	50000	signal	ECF_2	25927	0.0011719	0.000628404	0.013604	0.0087861	0.0342838	0.0357997	0.0442178	15	15
esup	50000	signal	ECF_3	25927	-0.052978	-0.0193191	0.16327	0.102191	0.626673	0.585182	-0.066208	2	2
esup	50000	signal	C_2	25927	0.000799795	0.00027226	0.00956625	0.0061269	0.0802279	0.0798414	-0.00481686	14	13
esup	50000	signal	D_2	25927	-0.011702	-0.0106751	0.0229883	0.0170517	0.207092	0.193448	-0.0658812	9	10
esup	50000	signal	L_2	25927	0.00345814	-0.00119814	0.0573852	0.0378829	0.407063	0.427381	0.0499142	4	4
esup	50000	signal	L_3	25927	0.00369082	0.00406937	0.0160022	0.0108565	0.0821137	0.077204	-0.0597914	13	14
esup	50000	signal	Q_w	25927	0.0179536	0.0126648	0.0488631	0.0352651	0.289722	0.2979	0.0282283	7	7
esup	50000	signal	ThrustMaj	25927	-0.0184363	-0.0123952	0.0561168	0.0412853	0.447112	0.441616	-0.0122928	3	3
esup	50000	background	τ_1^{WTA}	24073	-0.00112047	-0.00249973	0.0201787	0.0128053	0.28468	0.282601	-0.0073012	9	9
esup	50000	background	τ_2^{WTA}	24073	-0.00610593	-0.00387821	0.0197872	0.0132971	0.321393	0.326328	0.0153547	8	7
esup	50000	background	τ_3^{WTA}	24073	-0.00275205	-0.00137648	0.0198975	0.0127311	0.233614	0.235886	0.0097249	11	11
esup	50000	background	τ_4^{WTA}	24073	0.0153247	0.0130663	0.0454782	0.0323426	0.956137	0.960661	0.00473216	1	1
esup	50000	background	$\sqrt{d_{12}}$	24073	0.0125161	0.00757877	0.0372245	0.0257534	0.334703	0.325188	-0.0284281	7	8
esup	50000	background	$\sqrt{d_{23}}$	24073	0.0133032	0.0102022	0.0391454	0.02626	0.415126	0.408883	-0.0150386	4	4
esup	50000	background	ECF_1	24073	0.00665281	-0.0050611	0.0543947	0.0240488	0.161456	0.167748	0.038965	13	13
esup	50000	background	ECF_2	24073	0.00397672	0.00283502	0.0171544	0.0101886	0.0553029	0.0556475	0.00623047	15	15
esup	50000	background	ECF_3	24073	0.0303069	0.0313842	0.0886737	0.0603171	0.795229	0.769456	-0.0324098	2	2
esup	50000	background	C_2	24073	0.000226312	-0.00012821	0.01126	0.00660009	0.113048	0.113144	0.000856654	14	14
esup	50000	background	D_2	24073	-0.0113306	-0.00706074	0.025167	0.0162562	0.371285	0.373788	0.00674132	5	6
esup	50000	background	L_2	24073	-0.00886901	-0.00300932	0.0360537	0.0236436	0.418071	0.430315	0.0292866	3	3
esup	50000	background	L_3	24073	0.00273001	0.00229591	0.0249	0.0166543	0.197905	0.196365	-0.0077807	12	12
esup	50000	background	Q_w	24073	0.0173799	0.0112473	0.031043	0.0196894	0.252024	0.244814	-0.0286081	10	10

Systematic	Size	Subset	Feature	n	Mean ΔSHAP	Median ΔSHAP	Mean |ΔSHAP|	Median |ΔSHAP|	Nom. mean |SHAP|	Sys. mean |SHAP|	Rel. attr. drift	Nom. rank	Sys. rank
esup	50000	background	ThrustMaj	24073	-0.00461278	-0.00284562	0.0374899	0.0247984	0.370759	0.378633	0.0212371	6	5
esup	100000	all	τ_1^{WTA}	100000	-0.00443192	-0.00434025	0.0191639	0.0130233	0.239131	0.234035	-0.0213118	11	11
esup	100000	all	τ_2^{WTA}	100000	-0.00735514	-0.00517795	0.0194206	0.0137621	0.289774	0.293169	0.0117184	6	6
esup	100000	all	τ_3^{WTA}	100000	-0.00390298	-0.00215878	0.022488	0.0147564	0.265777	0.269441	0.0137863	9	9
esup	100000	all	τ_4^{WTA}	100000	0.0179069	0.0144962	0.0465714	0.0340235	0.914908	0.915549	0.000700995	1	1
esup	100000	all	$\sqrt{d_{12}}$	100000	0.0110216	0.00763958	0.0355621	0.0251217	0.263566	0.261896	-0.00633653	10	10
esup	100000	all	$\sqrt{d_{23}}$	100000	0.00683049	0.00525184	0.0421442	0.0294514	0.381965	0.377015	-0.012958	5	5
esup	100000	all	ECF_1	100000	-0.00520791	-0.00800674	0.0512201	0.0283191	0.153874	0.153756	-0.000762907	12	12
esup	100000	all	ECF_2	100000	0.00248428	0.00159834	0.015399	0.00939831	0.0442762	0.0452584	0.0221844	15	15
esup	100000	all	ECF_3	100000	-0.0128104	0.00986212	0.127205	0.0756971	0.706964	0.673663	-0.0471041	2	2
esup	100000	all	C_2	100000	0.000478312	6.12177e-05	0.0104347	0.00640296	0.0960899	0.0959988	-0.000948362	14	14
esup	100000	all	D_2	100000	-0.0115426	-0.00901648	0.0239895	0.0166137	0.286172	0.28033	-0.0204134	7	7
esup	100000	all	L_2	100000	-0.00248381	-0.00238393	0.0471862	0.0299634	0.413948	0.430607	0.0402431	3	3
esup	100000	all	L_3	100000	0.0032894	0.00334955	0.0202784	0.0130793	0.138227	0.134892	-0.0241263	13	13
esup	100000	all	Q_w	100000	0.0177597	0.0119417	0.0402534	0.0265733	0.271797	0.272716	0.00338268	8	8
esup	100000	all	ThrustMaj	100000	-0.0121745	-0.00685555	0.0470037	0.0323694	0.411536	0.412402	0.00210611	4	4
esup	100000	signal	τ_1^{WTA}	51851	-0.00739561	-0.00601216	0.0181823	0.013091	0.198255	0.190244	-0.0404051	10	11
esup	100000	signal	τ_2^{WTA}	51851	-0.00865511	-0.00655799	0.0190788	0.014121	0.26058	0.262693	0.00810844	8	8
esup	100000	signal	τ_3^{WTA}	51851	-0.00494974	-0.00327975	0.0250134	0.0170527	0.294808	0.299622	0.0163274	6	6
esup	100000	signal	τ_4^{WTA}	51851	0.0202038	0.0158541	0.0475725	0.0354851	0.879015	0.875911	-0.00353099	1	1
esup	100000	signal	$\sqrt{d_{12}}$	51851	0.00964228	0.00775301	0.0339011	0.0244609	0.197514	0.203156	0.0285645	11	9
esup	100000	signal	$\sqrt{d_{23}}$	51851	0.000961959	-0.000287414	0.0449685	0.0325209	0.351549	0.347763	-0.0107689	5	5
esup	100000	signal	ECF_1	51851	-0.0157303	-0.0118219	0.0487324	0.0328522	0.145992	0.139981	-0.0411721	12	12
esup	100000	signal	ECF_2	51851	0.00114394	0.000539167	0.0136528	0.00874949	0.0341046	0.0357019	0.0468342	15	15
esup	100000	signal	ECF_3	51851	-0.0525111	-0.0184747	0.16292	0.100925	0.624806	0.584159	-0.0650556	2	2
esup	100000	signal	C_2	51851	0.000771914	0.00029704	0.00956622	0.00615335	0.0800074	0.079708	-0.00374106	14	13
esup	100000	signal	D_2	51851	-0.0117442	-0.0106751	0.0229985	0.0169925	0.207536	0.193872	-0.0658407	9	10
esup	100000	signal	L_2	51851	0.00326186	-0.00109395	0.0575073	0.0380129	0.407436	0.428088	0.0506868	4	4
esup	100000	signal	L_3	51851	0.00362828	0.00397675	0.0160768	0.0108812	0.0818522	0.0769241	-0.060207	13	14
esup	100000	signal	Q_w	51851	0.0182386	0.0130228	0.0488023	0.0351582	0.289932	0.298148	0.0283356	7	7
esup	100000	signal	ThrustMaj	51851	-0.018611	-0.0126655	0.055957	0.0412667	0.44829	0.442619	-0.0126524	3	3
esup	100000	background	τ_1^{WTA}	48149	-0.00124036	-0.00262269	0.020221	0.012943	0.283157	0.281189	-0.00694705	9	9
esup	100000	background	τ_2^{WTA}	48149	-0.00595522	-0.0037545	0.0197887	0.0133459	0.32121	0.325986	0.0148683	8	7
esup	100000	background	τ_3^{WTA}	48149	-0.00277574	-0.0013648	0.0197684	0.0125363	0.234514	0.236938	0.0103368	11	11
esup	100000	background	τ_4^{WTA}	48149	0.0154334	0.0131942	0.0454933	0.0324218	0.953555	0.958226	0.00489786	1	1
esup	100000	background	$\sqrt{d_{12}}$	48149	0.0125071	0.00753541	0.0373509	0.0259081	0.334694	0.325151	-0.0285126	7	8
esup	100000	background	$\sqrt{d_{23}}$	48149	0.0131502	0.00999898	0.0391028	0.0263354	0.414708	0.408511	-0.0149415	4	4
esup	100000	background	ECF_1	48149	0.00612354	-0.00507588	0.0538991	0.0237906	0.162363	0.168589	0.0383491	13	13
esup	100000	background	ECF_2	48149	0.00392767	0.00282866	0.0172796	0.0102395	0.0552302	0.0555497	0.00578487	15	15
esup	100000	background	ECF_3	48149	0.0299427	0.0313201	0.088744	0.0603766	0.795439	0.770049	-0.0319192	2	2
esup	100000	background	C_2	48149	0.000162138	-0.000149064	0.0113699	0.00669482	0.113411	0.113542	0.00115985	14	14
esup	100000	background	D_2	48149	-0.0113255	-0.00710131	0.0250567	0.0161715	0.370859	0.373433	0.00693991	6	6
esup	100000	background	L_2	48149	-0.00867125	-0.00318803	0.0360716	0.0236498	0.420962	0.433315	0.029346	3	3
esup	100000	background	L_3	48149	0.00292446	0.00244758	0.024803	0.0165465	0.198937	0.197319	-0.00813306	12	12
esup	100000	background	Q_w	48149	0.017244	0.0113006	0.0310472	0.0197037	0.252271	0.245326	-0.0275308	10	10
esup	100000	background	ThrustMaj	48149	-0.00524318	-0.003167	0.037362	0.024838	0.37196	0.379871	0.0212695	5	5
esdown	25000	all	τ_1^{WTA}	25000	0.00548983	0.00495812	0.0198012	0.0135917	0.239349	0.244213	0.0203218	11	11
esdown	25000	all	τ_2^{WTA}	25000	0.00718007	0.00518979	0.0203258	0.0143468	0.288449	0.28478	-0.0127206	6	7
esdown	25000	all	τ_3^{WTA}	25000	0.00367754	0.00248356	0.0225276	0.014763	0.265475	0.26222	-0.0122599	9	10

Systematic	Size	Subset	Feature	n	Mean ΔSHAP	Median ΔSHAP	Mean |ΔSHAP|	Median |ΔSHAP|	Nom. mean |SHAP|	Sys. mean |SHAP|	Rel. attr. drift	Nom. rank	Sys. rank
esdown	25000	all	τ_4^{WTA}	25000	-0.0225386	-0.0194148	0.0495251	0.0362575	0.912488	0.908905	-0.00392586	1	1
esdown	25000	all	$\sqrt{d_{12}}$	25000	-0.0153461	-0.0112449	0.0386302	0.0277124	0.264725	0.264824	0.000374887	10	9
esdown	25000	all	$\sqrt{d_{23}}$	25000	-0.0149912	-0.0100492	0.0444465	0.0308905	0.381392	0.380984	-0.00107092	5	5
esdown	25000	all	ECF_1	25000	0.0098823	0.0106271	0.0504551	0.0293548	0.152646	0.156224	0.0234395	12	12
esdown	25000	all	ECF_2	25000	-0.00122314	-0.00140325	0.0163449	0.00985726	0.0444247	0.0448118	0.00871375	15	15
esdown	25000	all	ECF_3	25000	-0.0327011	-0.0263252	0.132458	0.0808765	0.710142	0.717062	0.00974425	2	2
esdown	25000	all	C_2	25000	-0.000137342	0.000109822	0.0108268	0.00670538	0.0961542	0.0964267	0.00283412	14	14
esdown	25000	all	D_2	25000	0.0116923	0.0093205	0.0247355	0.0173449	0.287497	0.29291	0.0188277	7	6
esdown	25000	all	L_2	25000	-0.00410622	-0.000422649	0.0478886	0.0305633	0.410395	0.389261	-0.0514981	3	4
esdown	25000	all	L_3	25000	-0.00527045	-0.00475534	0.0220958	0.0146313	0.139372	0.14392	0.0326331	13	13
esdown	25000	all	Q_w	25000	-0.0265144	-0.0169044	0.0440537	0.0286958	0.272915	0.266294	-0.0242626	8	8
esdown	25000	all	ThrustMaj	25000	0.00446142	0.00264595	0.0465374	0.0329176	0.409908	0.402456	-0.0181783	4	3
esdown	25000	signal	τ_1^{WTA}	12963	0.00776723	0.00669727	0.0185795	0.0134562	0.199079	0.206592	0.0377382	10	10
esdown	25000	signal	τ_2^{WTA}	12963	0.00807515	0.00638862	0.0198695	0.0148277	0.260017	0.257279	-0.0105324	8	8
esdown	25000	signal	τ_3^{WTA}	12963	0.00363772	0.002902	0.024616	0.0168098	0.295426	0.291406	-0.0136085	6	6
esdown	25000	signal	τ_4^{WTA}	12963	-0.0282572	-0.0242871	0.0512555	0.0386373	0.87599	0.873888	-0.00239871	1	1
esdown	25000	signal	$\sqrt{d_{12}}$	12963	-0.0171142	-0.0138306	0.0383262	0.0282646	0.197745	0.188206	-0.0482361	11	11
esdown	25000	signal	$\sqrt{d_{23}}$	12963	-0.0156298	-0.00965121	0.0481993	0.0347033	0.349892	0.342878	-0.0200452	5	5
esdown	25000	signal	ECF_1	12963	0.0170601	0.0133149	0.0490129	0.0331023	0.144653	0.154221	0.0661485	12	12
esdown	25000	signal	ECF_2	12963	-0.000569067	-0.000360582	0.013605	0.00912765	0.034188	0.0336373	-0.0161079	15	15
esdown	25000	signal	ECF_3	12963	-0.031625	-0.0140783	0.167195	0.108344	0.625003	0.611539	-0.0215431	2	2
esdown	25000	signal	C_2	12963	-0.000321605	-3.14713e-05	0.00977702	0.00646031	0.0797743	0.0803169	0.00680183	14	14
esdown	25000	signal	D_2	12963	0.0112653	0.0105805	0.0232269	0.0175788	0.20827	0.221039	0.0613105	9	9
esdown	25000	signal	L_2	12963	-0.0163431	-0.00645357	0.0578943	0.0384159	0.403392	0.374854	-0.0707449	4	4
esdown	25000	signal	L_3	12963	-0.00636042	-0.00581826	0.01851	0.012689	0.0822223	0.0894439	0.0878305	13	13
esdown	25000	signal	Q_w	12963	-0.0347208	-0.0266951	0.0557643	0.0407038	0.289396	0.269266	-0.0695591	7	7
esdown	25000	signal	ThrustMaj	12963	0.00339485	0.00108105	0.054266	0.0414612	0.446363	0.438703	-0.0171612	3	3
esdown	25000	background	τ_1^{WTA}	12037	0.00303724	0.00299107	0.0211169	0.0137612	0.282715	0.284725	0.00711075	9	9
esdown	25000	background	τ_2^{WTA}	12037	0.00621613	0.00400531	0.0208172	0.0139484	0.319067	0.314399	-0.0146316	8	8
esdown	25000	background	τ_3^{WTA}	12037	0.00372044	0.00222018	0.0202785	0.0126773	0.233219	0.230788	-0.0104236	11	11
esdown	25000	background	τ_4^{WTA}	12037	-0.01638	-0.0146014	0.0476617	0.0336933	0.951788	0.946616	-0.00543443	1	1
esdown	25000	background	$\sqrt{d_{12}}$	12037	-0.0134421	-0.00805992	0.0389576	0.0271171	0.336857	0.347334	0.0311021	7	7
esdown	25000	background	$\sqrt{d_{23}}$	12037	-0.0143034	-0.0105125	0.0404051	0.0273865	0.415312	0.422024	0.0161607	4	3
esdown	25000	background	ECF_1	12037	0.00215228	0.00806874	0.0520082	0.0258398	0.161256	0.15838	-0.0178372	13	13
esdown	25000	background	ECF_2	12037	-0.00192753	-0.0026983	0.0192955	0.0108105	0.0554486	0.0568459	0.025199	15	15
esdown	25000	background	ECF_3	12037	-0.03386	-0.035546	0.0950476	0.0647376	0.801836	0.830702	0.0360006	2	2
esdown	25000	background	C_2	12037	6.10956e-05	0.000286982	0.0119574	0.00701508	0.113794	0.113776	-0.000153407	14	14
esdown	25000	background	D_2	12037	0.0121521	0.0077296	0.0263602	0.0170687	0.37282	0.37031	-0.00673322	5	5
esdown	25000	background	L_2	12037	0.00907205	0.00378862	0.0371132	0.0241248	0.417938	0.404776	-0.0314908	3	4
esdown	25000	background	L_3	12037	-0.00409662	-0.00309901	0.0259574	0.0172921	0.200919	0.202587	0.0083036	12	12
esdown	25000	background	Q_w	12037	-0.0176768	-0.0115598	0.0314422	0.0199651	0.255166	0.263095	0.0310741	10	10
esdown	25000	background	ThrustMaj	12037	0.00561005	0.00348935	0.0382142	0.0253082	0.370652	0.36342	-0.0195111	6	6
esdown	50000	all	τ_1^{WTA}	50000	0.00544915	0.00497052	0.0198721	0.0135859	0.23978	0.244832	0.0210682	11	11
esdown	50000	all	τ_2^{WTA}	50000	0.00730408	0.0051336	0.0202522	0.0144204	0.289092	0.28537	-0.0128729	6	7
esdown	50000	all	τ_3^{WTA}	50000	0.00356458	0.00239408	0.0227156	0.0149031	0.265709	0.262351	-0.0126384	9	10
esdown	50000	all	τ_4^{WTA}	50000	-0.0233291	-0.019651	0.0497941	0.0365929	0.910554	0.907008	-0.00389401	1	1
esdown	50000	all	$\sqrt{d_{12}}$	50000	-0.0153948	-0.0112218	0.0386685	0.0277771	0.263809	0.263608	-0.00076243	10	9
esdown	50000	all	$\sqrt{d_{23}}$	50000	-0.014384	-0.00982974	0.0441975	0.0306493	0.380904	0.380689	-0.000562554	5	5
esdown	50000	all	ECF_1	50000	0.00954055	0.0107554	0.0504983	0.0293943	0.152917	0.156631	0.0242857	12	12

Systematic	Size	Subset	Feature	n	Mean ΔSHAP	Median ΔSHAP	Mean |ΔSHAP|	Median |ΔSHAP|	Nom. mean |SHAP|	Sys. mean |SHAP|	Rel. attr. drift	Nom. rank	Sys. rank
esdown	50000	all	ECF_2	50000	-0.00143357	-0.00146727	0.0162274	0.00983865	0.0443977	0.0446487	0.00565385	15	15
esdown	50000	all	ECF_3	50000	-0.0333813	-0.0264992	0.132161	0.0812522	0.7065	0.711993	0.00777552	2	2
esdown	50000	all	C_2	50000	-0.000136498	7.31051e-05	0.0107791	0.00668604	0.0957288	0.0960016	0.00284905	14	14
esdown	50000	all	D_2	50000	0.0120081	0.00947138	0.0247866	0.0173726	0.288376	0.293833	0.0189246	7	6
esdown	50000	all	L_2	50000	-0.00422468	-0.000689641	0.0480584	0.0304961	0.41236	0.390877	-0.052099	3	4
esdown	50000	all	L_3	50000	-0.00525858	-0.00483424	0.0222177	0.0146367	0.138349	0.142915	0.0330005	13	13
esdown	50000	all	Q_w	50000	-0.0261662	-0.0168796	0.0442678	0.0287176	0.272777	0.26623	-0.0240011	8	8
esdown	50000	all	ThrustMaj	50000	0.00464036	0.00275427	0.046718	0.032909	0.41146	0.403564	-0.0191904	4	3
esdown	50000	signal	τ_1^{WTA}	25927	0.00769006	0.00659946	0.0185624	0.0134839	0.199567	0.207069	0.0375914	10	10
esdown	50000	signal	τ_2^{WTA}	25927	0.00816798	0.00619876	0.0198572	0.0148861	0.260428	0.257596	-0.0108707	8	8
esdown	50000	signal	τ_3^{WTA}	25927	0.00345114	0.00279729	0.0249613	0.0169246	0.295006	0.291071	-0.0133372	6	6
esdown	50000	signal	τ_4^{WTA}	25927	-0.0288974	-0.0246541	0.0515973	0.0389872	0.874644	0.872663	-0.00226467	1	1
esdown	50000	signal	$\sqrt{d_{12}}$	25927	-0.0174792	-0.0141068	0.0386125	0.0284778	0.196997	0.187092	-0.0502847	11	11
esdown	50000	signal	$\sqrt{d_{23}}$	25927	-0.0148189	-0.00926033	0.0480251	0.034622	0.350083	0.343645	-0.0183884	5	5
esdown	50000	signal	ECF_1	25927	0.0166546	0.0134882	0.0491605	0.0330913	0.144833	0.154325	0.0655379	12	12
esdown	50000	signal	ECF_2	25927	-0.00079701	-0.000468791	0.0135842	0.00910597	0.0344003	0.0336873	-0.0207253	15	15
esdown	50000	signal	ECF_3	25927	-0.0335321	-0.0153356	0.167339	0.110229	0.625481	0.609972	-0.0247951	2	2
esdown	50000	signal	C_2	25927	-0.000373882	-0.00015305	0.00974878	0.00639381	0.0803986	0.0809436	0.00677876	14	14
esdown	50000	signal	D_2	25927	0.0115185	0.0106737	0.0233976	0.0176618	0.207771	0.220534	0.0614291	9	9
esdown	50000	signal	L_2	25927	-0.0168596	-0.00748497	0.0580064	0.0382079	0.404956	0.375733	-0.0721633	4	4
esdown	50000	signal	L_3	25927	-0.00656882	-0.00604007	0.0186978	0.0128753	0.0827248	0.0898128	0.0856819	13	13
esdown	50000	signal	Q_w	25927	-0.0344572	-0.0259801	0.0560514	0.0407192	0.289637	0.269433	-0.069756	7	7
esdown	50000	signal	ThrustMaj	25927	0.00342555	0.00117803	0.054437	0.0414562	0.446439	0.438389	-0.0180325	3	3
esdown	50000	background	τ_1^{WTA}	24073	0.00303566	0.00314736	0.0212826	0.0137193	0.28309	0.285506	0.00853656	9	9
esdown	50000	background	τ_2^{WTA}	24073	0.00637364	0.00406823	0.0206776	0.0139463	0.319964	0.315288	-0.0146141	8	8
esdown	50000	background	τ_3^{WTA}	24073	0.00368674	0.00211626	0.020297	0.0128492	0.234154	0.231417	-0.0116899	11	11
esdown	50000	background	τ_4^{WTA}	24073	-0.0173321	-0.0149391	0.0478521	0.0339202	0.949233	0.94399	-0.00552297	1	1
esdown	50000	background	$\sqrt{d_{12}}$	24073	-0.0131498	-0.00803709	0.0387288	0.0269999	0.335768	0.346017	0.0305249	7	7
esdown	50000	background	$\sqrt{d_{23}}$	24073	-0.0139156	-0.010448	0.0400752	0.0270003	0.414102	0.420583	0.0156508	4	3
esdown	50000	background	ECF_1	24073	0.0018786	0.00825118	0.0519391	0.0258905	0.161622	0.159115	-0.0155166	13	13
esdown	50000	background	ECF_2	24073	-0.00211916	-0.0027123	0.0190743	0.010774	0.0551649	0.0564539	0.023367	15	15
esdown	50000	background	ECF_3	24073	-0.0332189	-0.0352516	0.0942744	0.0648063	0.793759	0.821867	0.0354114	2	2
esdown	50000	background	C_2	24073	0.000119169	0.000328286	0.0118889	0.00705153	0.112241	0.11222	-0.000189183	14	14
esdown	50000	background	D_2	24073	0.0125355	0.00801361	0.0262825	0.0170362	0.375188	0.372779	-0.00642113	5	5
esdown	50000	background	L_2	24073	0.00938329	0.00381768	0.0373443	0.0241438	0.420332	0.407186	-0.0312762	3	4
esdown	50000	background	L_3	24073	-0.00384742	-0.00311358	0.0260088	0.0171657	0.198258	0.200104	0.00930996	12	12
esdown	50000	background	Q_w	24073	-0.0172367	-0.0116993	0.0315767	0.0204435	0.254618	0.262781	0.0320604	10	10
esdown	50000	background	ThrustMaj	24073	0.00594873	0.00374602	0.0384045	0.0254426	0.373784	0.366059	-0.0206671	6	6
esdown	100000	all	τ_1^{WTA}	100000	0.00534435	0.00486714	0.0198508	0.0135393	0.239359	0.244408	0.0210942	11	11
esdown	100000	all	τ_2^{WTA}	100000	0.00720541	0.00506014	0.0202738	0.014444	0.289806	0.285956	-0.0132858	6	7
esdown	100000	all	τ_3^{WTA}	100000	0.00346677	0.00234497	0.0228598	0.0149684	0.266534	0.263137	-0.0127462	9	9
esdown	100000	all	τ_4^{WTA}	100000	-0.0232354	-0.0193963	0.0498115	0.0366069	0.910574	0.907307	-0.00358751	1	1
esdown	100000	all	$\sqrt{d_{12}}$	100000	-0.0156328	-0.0113007	0.0389934	0.0279073	0.262793	0.262643	-0.000568505	10	10
esdown	100000	all	$\sqrt{d_{23}}$	100000	-0.0141728	-0.00979245	0.0442657	0.0306614	0.381419	0.380805	-0.00160959	5	5
esdown	100000	all	ECF_1	100000	0.00971306	0.0109327	0.0504284	0.0293521	0.15328	0.156883	0.0235102	12	12
esdown	100000	all	ECF_2	100000	-0.00146359	-0.00142894	0.0160902	0.00973163	0.0442419	0.0443198	0.00175933	15	15
esdown	100000	all	ECF_3	100000	-0.0328923	-0.0262338	0.132332	0.0812167	0.707233	0.71297	0.00811114	2	2
esdown	100000	all	C_2	100000	-8.99549e-05	6.1186e-05	0.0108126	0.00672704	0.0957582	0.0959462	0.00196282	14	14
esdown	100000	all	D_2	100000	0.0119329	0.00941069	0.0247884	0.0173063	0.286963	0.292475	0.0192094	7	6

Systematic	Size	Subset	Feature	n	Mean ΔSHAP	Median ΔSHAP	Mean |ΔSHAP|	Median |ΔSHAP|	Nom. mean |SHAP|	Sys. mean |SHAP|	Rel. attr. drift	Nom. rank	Sys. rank
esdown	100000	all	L_2	100000	-0.00401339	-0.000528201	0.0480699	0.0306167	0.412814	0.391654	-0.0512574	3	4
esdown	100000	all	L_3	100000	-0.00529391	-0.00491074	0.0221455	0.0145749	0.138091	0.142682	0.0332463	13	13
esdown	100000	all	Q_w	100000	-0.0265356	-0.017056	0.0446582	0.0288031	0.272023	0.265313	-0.0246656	8	8
esdown	100000	all	ThrustMaj	100000	0.00433202	0.0025569	0.0467127	0.0329132	0.410402	0.402607	-0.0189939	4	3
esdown	100000	signal	τ_1^{WTA}	51853	0.00767254	0.00655052	0.0186201	0.0135044	0.199249	0.20682	0.0379976	10	10
esdown	100000	signal	τ_2^{WTA}	51853	0.00804523	0.00616318	0.0198201	0.0148828	0.260987	0.258065	-0.0111965	8	8
esdown	100000	signal	τ_3^{WTA}	51853	0.00337349	0.00272498	0.0250432	0.0170673	0.295619	0.291542	-0.0137899	6	6
esdown	100000	signal	τ_4^{WTA}	51853	-0.0287042	-0.0241066	0.0516104	0.0388809	0.876644	0.875072	-0.00179369	1	1
esdown	100000	signal	$\sqrt{d_{12}}$	51853	-0.0179582	-0.0144712	0.0389669	0.02857	0.196815	0.186723	-0.0512773	11	11
esdown	100000	signal	$\sqrt{d_{23}}$	51853	-0.0145739	-0.00937439	0.048091	0.0346517	0.350571	0.343429	-0.0203709	5	5
esdown	100000	signal	ECF_1	51853	0.0170519	0.0136396	0.0489855	0.0330192	0.145377	0.155069	0.0666689	12	12
esdown	100000	signal	ECF_2	51853	-0.000838902	-0.000469965	0.0134496	0.00895782	0.0343047	0.0334075	-0.0261551	15	15
esdown	100000	signal	ECF_3	51853	-0.0331425	-0.0153742	0.167648	0.109858	0.62518	0.610168	-0.0240124	2	2
esdown	100000	signal	C_2	51853	-0.000398099	-0.000163279	0.00977628	0.00641411	0.079899	0.080366	0.00584547	14	14
esdown	100000	signal	D_2	51853	0.0114525	0.0107328	0.0234113	0.0176016	0.207521	0.220392	0.0620253	9	9
esdown	100000	signal	L_2	51853	-0.016591	-0.00715986	0.0579745	0.0384137	0.406232	0.377578	-0.0705357	4	4
esdown	100000	signal	L_3	51853	-0.00659795	-0.00614977	0.018527	0.0127346	0.0822517	0.0891937	0.0843994	13	13
esdown	100000	signal	Q_w	51853	-0.0347862	-0.0260493	0.056534	0.0410207	0.288838	0.268278	-0.0711832	7	7
esdown	100000	signal	ThrustMaj	51853	0.00295539	0.0008533	0.0542549	0.0414427	0.445973	0.43795	-0.0179895	3	3
esdown	100000	background	τ_1^{WTA}	48147	0.00283696	0.00301856	0.0211763	0.0135757	0.282555	0.28489	0.00826286	9	9
esdown	100000	background	τ_2^{WTA}	48147	0.00630096	0.00398201	0.0207624	0.013987	0.320846	0.315998	-0.0151097	8	8
esdown	100000	background	τ_3^{WTA}	48147	0.00356723	0.00207207	0.0205082	0.0128736	0.235213	0.232549	-0.0113271	11	11
esdown	100000	background	τ_4^{WTA}	48147	-0.0173456	-0.0148017	0.0478742	0.0341155	0.94713	0.942027	-0.00538772	1	1
esdown	100000	background	$\sqrt{d_{12}}$	48147	-0.0131284	-0.00791937	0.0390219	0.027162	0.333853	0.344414	0.0316328	7	7
esdown	100000	background	$\sqrt{d_{23}}$	48147	-0.0137408	-0.010205	0.0401459	0.0268919	0.414641	0.421054	0.0154653	4	3
esdown	100000	background	ECF_1	48147	0.00180934	0.00845207	0.0519824	0.0257647	0.16179	0.158837	-0.0182502	13	13
esdown	100000	background	ECF_2	48147	-0.00213637	-0.00263203	0.018934	0.0107668	0.0549448	0.0560721	0.0205169	15	15
esdown	100000	background	ECF_3	48147	-0.0326229	-0.0343883	0.0942976	0.06462	0.795605	0.823693	0.0353034	2	2
esdown	100000	background	C_2	48147	0.000241908	0.000307426	0.0119288	0.00712524	0.112839	0.112726	-0.00100495	14	14
esdown	100000	background	D_2	48147	0.0124503	0.0078131	0.0262715	0.0169691	0.372517	0.370102	-0.00648444	5	5
esdown	100000	background	L_2	48147	0.00953237	0.00381666	0.0374029	0.0241723	0.419905	0.406817	-0.0311691	3	4
esdown	100000	background	L_3	48147	-0.00388949	-0.00313222	0.0260424	0.0172283	0.198228	0.200289	0.0103945	12	12
esdown	100000	background	Q_w	48147	-0.0176499	-0.0118033	0.0318683	0.0203964	0.253911	0.262117	0.0323205	10	10
esdown	100000	background	ThrustMaj	48147	0.00581462	0.0035831	0.03859	0.0254574	0.372097	0.364547	-0.0202899	6	6
cpos	25000	all	τ_1^{WTA}	25000	3.42933e-05	0	0.00294052	3.18326e-05	0.239263	0.239344	0.000337181	11	11
cpos	25000	all	τ_2^{WTA}	25000	0.000205814	0	0.00453562	3.68655e-05	0.290105	0.290034	-0.000244393	6	6
cpos	25000	all	τ_3^{WTA}	25000	-4.36753e-05	0	0.00555819	6.50883e-05	0.266625	0.266693	0.000253173	9	9
cpos	25000	all	τ_4^{WTA}	25000	-0.000414701	0	0.0131249	0.000388533	0.907388	0.907078	-0.00034217	1	1
cpos	25000	all	$\sqrt{d_{12}}$	25000	-9.57921e-06	0	0.00543589	0.000150882	0.264572	0.264453	-0.000450462	10	10
cpos	25000	all	$\sqrt{d_{23}}$	25000	-3.68057e-05	0	0.00801595	0.000283554	0.381048	0.380997	-0.000134602	5	5
cpos	25000	all	ECF_1	25000	-1.08697e-05	0	0.00319144	7.10189e-05	0.15373	0.153702	-0.000181551	12	12
cpos	25000	all	ECF_2	25000	-4.37941e-05	0	0.00166235	1.27628e-05	0.0442467	0.044218	-0.000648039	15	15
cpos	25000	all	ECF_3	25000	5.28998e-05	0	0.00843733	0.000395387	0.715047	0.715101	7.53553e-05	2	2
cpos	25000	all	C_2	25000	2.08995e-05	0	0.00214031	3.7428e-05	0.0974392	0.0974762	0.000379108	14	14
cpos	25000	all	D_2	25000	-0.000126296	0	0.00402753	0.000115213	0.285922	0.285959	0.00013029	7	7
cpos	25000	all	L_2	25000	0.0003422	0	0.00624741	0.000269104	0.415616	0.415412	-0.000488751	3	3
cpos	25000	all	L_3	25000	-6.88588e-05	0	0.00348623	6.9987e-05	0.139558	0.139489	-0.00049906	13	13
cpos	25000	all	Q_w	25000	0.000337506	0	0.00603792	0.000174262	0.27107	0.271326	0.000943313	8	8
cpos	25000	all	ThrustMaj	25000	0.000369438	0	0.00737067	0.000285715	0.411984	0.411954	-7.23385e-05	4	4

Systematic	Size	Subset	Feature	n	Mean ΔSHAP	Median ΔSHAP	Mean |ΔSHAP|	Median |ΔSHAP|	Nom. mean |SHAP|	Sys. mean |SHAP|	Rel. attr. drift	Nom. rank	Sys. rank
cpos	25000	signal	τ_1^{WTA}	12963	-4.65354e-05	0	0.00230213	3.57628e-07	0.199802	0.199744	-0.000292054	10	10
cpos	25000	signal	τ_2^{WTA}	12963	0.000193769	0	0.00348952	1.53482e-06	0.261553	0.261412	-0.000540435	8	8
cpos	25000	signal	τ_3^{WTA}	12963	-7.30481e-05	0	0.00504036	9.0003e-06	0.296385	0.296514	0.000437204	6	6
cpos	25000	signal	τ_4^{WTA}	12963	-0.000823612	0	0.01111	8.51452e-05	0.876649	0.876492	-0.000178954	1	1
cpos	25000	signal	$\sqrt{d_{12}}$	12963	4.19361e-05	0	0.00405804	3.2872e-05	0.195527	0.195363	-0.000841285	11	11
cpos	25000	signal	$\sqrt{d_{23}}$	12963	0.000158237	0	0.00615944	7.48932e-05	0.349928	0.349846	-0.000232591	5	5
cpos	25000	signal	ECF_1	12963	-4.31137e-05	0	0.00221323	1.13249e-05	0.144886	0.144748	-0.000953395	12	12
cpos	25000	signal	ECF_2	12963	-8.52821e-06	0	0.00108731	1.49012e-08	0.0339882	0.0339349	-0.00156955	15	15
cpos	25000	signal	ECF_3	12963	7.99052e-05	0	0.00718308	0.00011456	0.625401	0.625362	-6.36645e-05	2	2
cpos	25000	signal	C_2	12963	-3.0085e-05	0	0.00164417	3.76254e-06	0.0800476	0.0801223	0.000932723	14	14
cpos	25000	signal	D_2	12963	-9.14908e-05	0	0.00274349	2.23368e-05	0.207138	0.20712	-8.70453e-05	9	9
cpos	25000	signal	L_2	12963	8.89963e-05	0	0.00532383	7.04527e-05	0.406923	0.40679	-0.000328693	4	4
cpos	25000	signal	L_3	12963	-0.000113422	0	0.00219947	1.20979e-05	0.081696	0.0816449	-0.000626353	13	13
cpos	25000	signal	Q_w	12963	0.000381205	0	0.0052849	4.46439e-05	0.28755	0.287619	0.000239725	7	7
cpos	25000	signal	ThrustMaj	12963	0.000248874	0	0.00651146	7.07507e-05	0.449377	0.44955	0.000384519	3	3
cpos	25000	background	τ_1^{WTA}	12037	0.00012134	0	0.00362801	0.000142813	0.281762	0.281989	0.000806823	9	9
cpos	25000	background	τ_2^{WTA}	12037	0.000218785	0	0.0056622	0.000149101	0.320853	0.320857	1.21679e-05	8	8
cpos	25000	background	τ_3^{WTA}	12037	-1.20428e-05	0	0.00611585	0.00019446	0.234576	0.234576	7.62286e-07	11	11
cpos	25000	background	τ_4^{WTA}	12037	2.56668e-05	0	0.0152949	0.00100687	0.940496	0.940019	-0.000506372	1	1
cpos	25000	background	$\sqrt{d_{12}}$	12037	-6.50576e-05	0	0.00691974	0.000408798	0.338931	0.338857	-0.000216045	7	7
cpos	25000	background	$\sqrt{d_{23}}$	12037	-0.000246853	0	0.0100153	0.000774622	0.414559	0.414541	-4.48589e-05	4	4
cpos	25000	background	ECF_1	12037	2.38547e-05	0	0.00424491	0.000198483	0.163254	0.163345	0.000557878	13	13
cpos	25000	background	ECF_2	12037	-8.1773e-05	0	0.00228162	4.88572e-05	0.0552946	0.0552923	-4.12989e-05	15	15
cpos	25000	background	ECF_3	12037	2.38169e-05	0	0.00978807	0.00093329	0.811587	0.811741	0.000189554	2	2
cpos	25000	background	C_2	12037	7.58061e-05	0	0.00267461	0.000130709	0.116169	0.116165	-3.33507e-05	14	14
cpos	25000	background	D_2	12037	-0.000163779	0	0.00541035	0.000307143	0.370768	0.370864	0.000260029	6	6
cpos	25000	background	L_2	12037	0.000614883	0	0.00724205	0.000640824	0.424974	0.424697	-0.000652815	3	3
cpos	25000	background	L_3	12037	-2.0867e-05	0	0.00487197	0.000210514	0.201873	0.201783	-0.000442741	12	12
cpos	25000	background	Q_w	12037	0.000290445	0	0.00684887	0.000400126	0.253325	0.25378	0.00179619	10	10
cpos	25000	background	ThrustMaj	12037	0.000499277	0	0.00829597	0.000673801	0.371716	0.371466	-0.000671145	5	5
cpos	50000	all	τ_1^{WTA}	50000	2.57342e-05	0	0.00293048	3.60459e-05	0.238936	0.238978	0.000174496	11	11
cpos	50000	all	τ_2^{WTA}	50000	0.000204392	0	0.00451146	3.85791e-05	0.289775	0.289672	-0.000358111	6	6
cpos	50000	all	τ_3^{WTA}	50000	6.38116e-05	0	0.00547029	6.7044e-05	0.266565	0.266544	-7.6584e-05	9	9
cpos	50000	all	τ_4^{WTA}	50000	-0.000456742	0	0.0129519	0.000405431	0.907668	0.907559	-0.000119844	1	1
cpos	50000	all	$\sqrt{d_{12}}$	50000	6.43722e-05	0	0.00554501	0.000164002	0.263493	0.263386	-0.00040582	10	10
cpos	50000	all	$\sqrt{d_{23}}$	50000	3.89783e-05	0	0.00811507	0.000299573	0.381165	0.381169	9.69525e-06	5	5
cpos	50000	all	ECF_1	50000	-3.66697e-05	0	0.00320373	7.22678e-05	0.15381	0.153766	-0.000285895	12	12
cpos	50000	all	ECF_2	50000	-4.22326e-06	0	0.00165457	1.23531e-05	0.0443042	0.0442828	-0.000483485	15	15
cpos	50000	all	ECF_3	50000	-6.79538e-06	0	0.00860197	0.000423983	0.716062	0.715985	-0.000106963	2	2
cpos	50000	all	C_2	50000	-1.17013e-05	0	0.00212609	3.79048e-05	0.0964756	0.0965221	0.000482132	14	14
cpos	50000	all	D_2	50000	-7.72105e-05	0	0.00401788	0.000123908	0.285454	0.285436	-6.03452e-05	7	7
cpos	50000	all	L_2	50000	0.000327696	0	0.00636399	0.000286147	0.413679	0.413439	-0.000580372	3	3
cpos	50000	all	L_3	50000	-5.86798e-05	0	0.0035144	7.19e-05	0.139335	0.139235	-0.000718778	13	13
cpos	50000	all	Q_w	50000	0.000259844	0	0.00602829	0.000183668	0.27183	0.271949	0.00043975	8	8
cpos	50000	all	ThrustMaj	50000	0.00034003	0	0.00741499	0.00029403	0.411787	0.411836	0.000116883	4	4
cpos	50000	signal	τ_1^{WTA}	25926	-3.64379e-05	0	0.00232547	1.43982e-06	0.199358	0.199304	-0.00027342	10	10
cpos	50000	signal	τ_2^{WTA}	25926	0.000197408	0	0.00339655	3.13018e-06	0.260565	0.260448	-0.000450411	8	8
cpos	50000	signal	τ_3^{WTA}	25926	-8.69507e-06	0	0.00495289	9.80496e-06	0.295624	0.295591	-0.00011049	6	6
cpos	50000	signal	τ_4^{WTA}	25926	-0.000854419	0	0.0108285	9.98452e-05	0.878199	0.878192	-7.94096e-06	1	1

Systematic	Size	Subset	Feature	n	Mean ΔSHAP	Median ΔSHAP	Mean |ΔSHAP|	Median |ΔSHAP|	Nom. mean |SHAP|	Sys. mean |SHAP|	Rel. attr. drift	Nom. rank	Sys. rank
cpos	50000	signal	$\sqrt{d_{12}}$	25926	8.44337e-05	0	0.00406253	3.77018e-05	0.195733	0.195636	-0.000493933	11	11
cpos	50000	signal	$\sqrt{d_{23}}$	25926	0.000311541	0	0.00619883	8.71867e-05	0.350488	0.350459	-8.46057e-05	5	5
cpos	50000	signal	ECF_1	25926	-5.35907e-05	0	0.00221538	1.45212e-05	0.145477	0.1454	-0.000530585	12	12
cpos	50000	signal	ECF_2	25926	-2.35278e-05	0	0.0010758	1.63913e-07	0.0342281	0.0341629	-0.0019052	15	15
cpos	50000	signal	ECF_3	25926	-0.000106259	0	0.00719283	0.000130504	0.628987	0.628887	-0.000158823	2	2
cpos	50000	signal	C_2	25926	-5.40473e-05	0	0.00162777	4.62495e-06	0.0793848	0.0794512	0.000836333	14	14
cpos	50000	signal	D_2	25926	-3.92414e-05	0	0.00276067	2.60966e-05	0.208039	0.207996	-0.000205999	9	9
cpos	50000	signal	L_2	25926	4.38556e-05	0	0.00541999	8.4132e-05	0.405921	0.405799	-0.000300798	4	4
cpos	50000	signal	L_3	25926	-9.19081e-05	0	0.00218118	1.37612e-05	0.081663	0.0816044	-0.000716748	13	13
cpos	50000	signal	Q_w	25926	0.00029302	0	0.00527156	5.42849e-05	0.288338	0.288369	0.000107287	7	7
cpos	50000	signal	ThrustMaj	25926	3.59521e-05	0	0.00642838	7.83056e-05	0.449833	0.450021	0.000418381	3	3
cpos	50000	background	τ_1^{WTA}	24074	9.26892e-05	0	0.00358203	0.000145487	0.28156	0.281705	0.000513887	9	9
cpos	50000	background	τ_2^{WTA}	24074	0.000211914	0	0.00571215	0.000141956	0.32123	0.32114	-0.000281852	8	8
cpos	50000	background	τ_3^{WTA}	24074	0.000141896	0	0.00602749	0.000188351	0.235273	0.235264	-3.82546e-05	11	11
cpos	50000	background	τ_4^{WTA}	24074	-2.8471e-05	0	0.0152386	0.00101373	0.939414	0.939194	-0.000233999	1	1
cpos	50000	background	$\sqrt{d_{12}}$	24074	4.27673e-05	0	0.00714155	0.000422537	0.336466	0.336349	-0.000347567	7	7
cpos	50000	background	$\sqrt{d_{23}}$	24074	-0.000254552	0	0.0101787	0.000755869	0.4142	0.414244	0.000104689	4	4
cpos	50000	background	ECF_1	24074	-1.84471e-05	0	0.00426812	0.000198789	0.162783	0.162775	-5.08961e-05	13	13
cpos	50000	background	ECF_2	24074	1.65663e-05	0	0.00227785	4.75952e-05	0.0551557	0.0551816	0.000469682	15	15
cpos	50000	background	ECF_3	24074	0.00010032	0	0.0101195	0.000954002	0.809836	0.809783	-6.602e-05	2	2
cpos	50000	background	C_2	24074	3.39024e-05	0	0.00266274	0.000125099	0.114881	0.114906	0.000222322	14	14
cpos	50000	background	D_2	24074	-0.000118101	0	0.0053718	0.000322171	0.368824	0.368838	3.64424e-05	6	6
cpos	50000	background	L_2	24074	0.000633371	0	0.00738061	0.000664087	0.422037	0.421667	-0.000877537	3	3
cpos	50000	background	L_3	24074	-2.28953e-05	0	0.00495019	0.000207413	0.201442	0.201297	-0.000718126	12	12
cpos	50000	background	Q_w	24074	0.000224117	0	0.00684324	0.000402324	0.254051	0.254267	0.000850018	10	10
cpos	50000	background	ThrustMaj	24074	0.000667501	0	0.00847751	0.000677586	0.370813	0.370712	-0.000270607	5	5
cpos	100000	all	τ_1^{WTA}	100000	2.10144e-05	0	0.00288102	3.61837e-05	0.23918	0.239186	2.42974e-05	11	11
cpos	100000	all	τ_2^{WTA}	100000	0.000154604	0	0.00448118	3.66122e-05	0.289843	0.289711	-0.000455811	6	6
cpos	100000	all	τ_3^{WTA}	100000	6.09334e-05	0	0.0055386	6.38273e-05	0.266878	0.266916	0.00014171	9	9
cpos	100000	all	τ_4^{WTA}	100000	-0.00052115	0	0.0128198	0.000397295	0.909541	0.909397	-0.000158589	1	1
cpos	100000	all	$\sqrt{d_{12}}$	100000	9.16216e-06	0	0.00550773	0.000163957	0.263711	0.263627	-0.000319031	10	10
cpos	100000	all	$\sqrt{d_{23}}$	100000	0.000109435	0	0.00806241	0.000295162	0.38138	0.38141	7.71275e-05	5	5
cpos	100000	all	ECF_1	100000	-2.31914e-05	0	0.00318026	7.07805e-05	0.153813	0.153805	-5.38643e-05	12	12
cpos	100000	all	ECF_2	100000	-3.38938e-06	0	0.00164582	1.20346e-05	0.0444089	0.0443927	-0.000364988	15	15
cpos	100000	all	ECF_3	100000	-9.78237e-05	0	0.00849404	0.000418544	0.711873	0.711793	-0.000112951	2	2
cpos	100000	all	C_2	100000	-1.25456e-06	0	0.00210311	3.66867e-05	0.0961373	0.0961621	0.000258538	14	14
cpos	100000	all	D_2	100000	-2.65281e-05	0	0.00396589	0.000119202	0.285994	0.285943	-0.000178713	7	7
cpos	100000	all	L_2	100000	0.000259015	0	0.00631768	0.00028877	0.414281	0.414111	-0.000411626	3	3
cpos	100000	all	L_3	100000	-2.99949e-05	0	0.00345842	7.07097e-05	0.13873	0.13867	-0.000433512	13	13
cpos	100000	all	Q_w	100000	0.000246391	0	0.00585732	0.000181288	0.27118	0.271312	0.000485752	8	8
cpos	100000	all	ThrustMaj	100000	0.000216746	0	0.00730477	0.000288695	0.411233	0.411196	-9.14579e-05	4	4
cpos	100000	signal	τ_1^{WTA}	51852	-3.33574e-05	0	0.00226642	1.04308e-06	0.199154	0.199116	-0.00018945	10	10
cpos	100000	signal	τ_2^{WTA}	51852	0.000182688	0	0.00337933	2.27243e-06	0.26106	0.260929	-0.000503555	8	8
cpos	100000	signal	τ_3^{WTA}	51852	-3.025e-05	0	0.00509897	9.37283e-06	0.296904	0.296885	-6.25349e-05	6	6
cpos	100000	signal	τ_4^{WTA}	51852	-0.00081385	0	0.0107635	9.68166e-05	0.877911	0.877768	-0.000162741	1	1
cpos	100000	signal	$\sqrt{d_{12}}$	51852	1.67801e-05	0	0.00397092	3.57628e-05	0.196782	0.19672	-0.00031312	11	11
cpos	100000	signal	$\sqrt{d_{23}}$	51852	0.0002832	0	0.00606789	7.96318e-05	0.350749	0.350844	0.000268837	5	5
cpos	100000	signal	ECF_1	51852	-2.81989e-05	0	0.00216523	1.21612e-05	0.145068	0.145006	-0.000432033	12	12
cpos	100000	signal	ECF_2	51852	-2.61724e-05	0	0.0010738	6.33299e-08	0.0342996	0.0342522	-0.00138098	15	15

Systematic	Size	Subset	Feature	n	Mean ΔSHAP	Median ΔSHAP	Mean |ΔSHAP|	Median |ΔSHAP|	Nom. mean |SHAP|	Sys. mean |SHAP|	Rel. attr. drift	Nom. rank	Sys. rank
cpos	100000	signal	ECF_3	51852	-0.000252365	0	0.00703136	0.000121385	0.626462	0.626303	-0.000253942	2	2
cpos	100000	signal	C_2	51852	-3.68573e-05	0	0.00159958	3.76254e-06	0.0798236	0.0798782	0.000683048	14	14
cpos	100000	signal	D_2	51852	-5.31443e-05	0	0.00271412	2.48253e-05	0.207442	0.2074	-0.000203503	9	9
cpos	100000	signal	L_2	51852	0.000179278	0	0.00540132	7.59363e-05	0.406223	0.406185	-9.35398e-05	4	4
cpos	100000	signal	L_3	51852	-4.4896e-05	0	0.00214467	1.22823e-05	0.0817971	0.0817803	-0.000205581	13	13
cpos	100000	signal	Q_w	51852	0.000264986	0	0.00510033	5.00083e-05	0.287939	0.288131	0.000665415	7	7
cpos	100000	signal	ThrustMaj	51852	6.73802e-06	0	0.00630085	7.34627e-05	0.448183	0.448263	0.000178342	3	3
cpos	100000	background	τ_1^{WTA}	48148	7.95689e-05	0	0.00354291	0.000153966	0.282287	0.282339	0.000182116	9	9
cpos	100000	background	τ_2^{WTA}	48148	0.00012436	0	0.0056678	0.000141561	0.320838	0.320711	-0.000394593	8	8
cpos	100000	background	τ_3^{WTA}	48148	0.000159132	0	0.00601205	0.000182901	0.23454	0.23464	0.000427645	11	11
cpos	100000	background	τ_4^{WTA}	48148	-0.000205932	0	0.0150342	0.00101137	0.943602	0.943465	-0.000144905	1	1
cpos	100000	background	$\sqrt{d_{12}}$	48148	9.58215e-07	0	0.00716276	0.000432909	0.335795	0.335685	-0.000327405	7	7
cpos	100000	background	$\sqrt{d_{23}}$	48148	-7.76986e-05	0	0.0102104	0.000743613	0.414376	0.414331	-0.000107162	4	4
cpos	100000	background	ECF_1	48148	-1.77987e-05	0	0.00427338	0.000207875	0.163231	0.163282	0.000310382	13	13
cpos	100000	background	ECF_2	48148	2.11463e-05	0	0.00226184	4.98742e-05	0.0552959	0.0553128	0.000305186	15	15
cpos	100000	background	ECF_3	48148	6.86069e-05	0	0.0100692	0.000971451	0.803849	0.803852	4.67139e-06	2	2
cpos	100000	background	C_2	48148	3.70871e-05	0	0.00264538	0.000130637	0.113707	0.113698	-7.36496e-05	14	14
cpos	100000	background	D_2	48148	2.13572e-06	0	0.00531395	0.000322342	0.37059	0.370529	-0.000163089	6	6
cpos	100000	background	L_2	48148	0.000344886	0	0.00730454	0.000675935	0.422957	0.422652	-0.000720261	3	3
cpos	100000	background	L_3	48148	-1.39474e-05	0	0.00487324	0.000220345	0.200043	0.199937	-0.000529473	12	12
cpos	100000	background	Q_w	48148	0.000226365	0	0.00667254	0.000401363	0.253131	0.253199	0.000270908	10	10
cpos	100000	background	ThrustMaj	48148	0.00044291	0	0.00838591	0.000678167	0.371439	0.371277	-0.00043768	5	5
cer	25000	all	τ_1^{WTA}	25000	0.000181344	1.86265e-06	0.0195862	0.0121055	0.238806	0.238636	-0.000710034	11	11
cer	25000	all	τ_2^{WTA}	25000	5.8264e-05	6.15418e-06	0.0238799	0.0161765	0.290322	0.290047	-0.000945124	6	6
cer	25000	all	τ_3^{WTA}	25000	-0.00018848	-5.44321e-05	0.025265	0.0151189	0.267398	0.267735	0.00126132	9	9
cer	25000	all	τ_4^{WTA}	25000	0.00119553	0.001414	0.0586429	0.0409945	0.917446	0.917833	0.000422617	1	1
cer	25000	all	$\sqrt{d_{12}}$	25000	-0.00039206	-0.000411381	0.0316468	0.0194356	0.264698	0.264712	5.3255e-05	10	10
cer	25000	all	$\sqrt{d_{23}}$	25000	-0.00105627	-0.000336617	0.0382733	0.0231987	0.381434	0.381353	-0.000214395	5	5
cer	25000	all	ECF_1	25000	0.000680982	0.000263162	0.0274755	0.0137624	0.153196	0.153686	0.00320024	12	12
cer	25000	all	ECF_2	25000	0.000147189	-3.19164e-05	0.0109021	0.00617469	0.0441951	0.044228	0.000744045	15	15
cer	25000	all	ECF_3	25000	-0.00426006	-0.000997424	0.061661	0.0372742	0.707695	0.704872	-0.00398908	2	2
cer	25000	all	C_2	25000	0.000112989	2.3867e-05	0.0107625	0.00652738	0.0969508	0.0968397	-0.00114628	14	14
cer	25000	all	D_2	25000	0.000194329	0.000213735	0.023789	0.0138367	0.283985	0.284287	0.00106413	7	7
cer	25000	all	L_2	25000	-0.00121294	-0.000669625	0.0367679	0.0250681	0.414871	0.414113	-0.00182706	3	3
cer	25000	all	L_3	25000	-6.82029e-05	5.15096e-05	0.020106	0.0118546	0.138166	0.138563	0.00287527	13	13
cer	25000	all	Q_w	25000	-0.000867031	-0.000107184	0.0331201	0.0180477	0.272751	0.272253	-0.00182441	8	8
cer	25000	all	ThrustMaj	25000	-0.000904688	-0.000527658	0.0394743	0.0265879	0.410391	0.409189	-0.00292867	4	4
cer	25000	signal	τ_1^{WTA}	12964	3.77335e-05	4.372e-05	0.0182392	0.0112914	0.198751	0.198692	-0.000295398	11	10
cer	25000	signal	τ_2^{WTA}	12964	-0.000454372	-0.000186215	0.0235228	0.0162596	0.26025	0.260011	-0.000916115	8	8
cer	25000	signal	τ_3^{WTA}	12964	-0.000708295	-0.000435695	0.0269788	0.0162866	0.298127	0.298724	0.0020041	6	6
cer	25000	signal	τ_4^{WTA}	12964	0.000929266	0.00129583	0.0573109	0.0405031	0.881569	0.88299	0.00161106	1	1
cer	25000	signal	$\sqrt{d_{12}}$	12964	-0.00129329	-0.00077017	0.0280172	0.0171252	0.199111	0.198624	-0.00244609	10	11
cer	25000	signal	$\sqrt{d_{23}}$	12964	-0.00154506	-0.00079985	0.0355023	0.0227038	0.352496	0.351317	-0.00334289	5	5
cer	25000	signal	ECF_1	12964	-0.00033189	2.0586e-05	0.0233308	0.0130173	0.144926	0.145666	0.00510568	12	12
cer	25000	signal	ECF_2	12964	3.38874e-05	-7.32351e-05	0.00895104	0.00532294	0.0344277	0.0345036	0.00220351	15	15
cer	25000	signal	ECF_3	12964	-0.00712269	-0.00173044	0.0670717	0.0386957	0.62472	0.619136	-0.0089384	2	2
cer	25000	signal	C_2	12964	3.38814e-05	6.08023e-05	0.00943925	0.00587116	0.0810565	0.0808274	-0.00282667	14	14
cer	25000	signal	D_2	12964	-0.000144957	5.57303e-06	0.019452	0.0121417	0.207089	0.207265	0.000851591	9	9
cer	25000	signal	L_2	12964	-0.00240022	-0.00152725	0.0381942	0.026656	0.408094	0.40652	-0.00385844	4	4

Systematic	Size	Subset	Feature	n	Mean ΔSHAP	Median ΔSHAP	Mean |ΔSHAP|	Median |ΔSHAP|	Nom. mean |SHAP|	Sys. mean |SHAP|	Rel. attr. drift	Nom. rank	Sys. rank
cer	25000	signal	L_3	12964	-0.00033221	-1.98837e-07	0.0146288	0.00900548	0.0823522	0.0827794	0.0051884	13	13
cer	25000	signal	Q_w	12964	-0.00179187	-0.000418596	0.0353821	0.0200886	0.291381	0.290171	-0.00414979	7	7
cer	25000	signal	ThrustMaj	12964	-0.00204435	-0.00105298	0.0420839	0.0289901	0.445813	0.4438	-0.00451547	3	3
cer	25000	background	τ_1^{WTA}	12036	0.000336028	-2.46614e-05	0.0210371	0.0131029	0.281949	0.28166	-0.00102467	9	9
cer	25000	background	τ_2^{WTA}	12036	0.000610425	0.000158407	0.0242646	0.0161042	0.322716	0.322399	-0.000983603	8	8
cer	25000	background	τ_3^{WTA}	12036	0.000371414	0.000267304	0.023419	0.0139697	0.234301	0.234356	0.000237668	11	11
cer	25000	background	τ_4^{WTA}	12036	0.00148232	0.00158322	0.0600776	0.0414101	0.956093	0.955361	-0.000765808	1	1
cer	25000	background	$\sqrt{d_{12}}$	12036	0.000578657	8.73841e-05	0.0355562	0.0224312	0.33534	0.335894	0.00165026	7	7
cer	25000	background	$\sqrt{d_{23}}$	12036	-0.000529791	7.0855e-05	0.041258	0.0236428	0.412604	0.413705	0.00266947	4	4
cer	25000	background	ECF_1	12036	0.00177195	0.000590203	0.0319398	0.0147187	0.162104	0.162324	0.00135366	13	13
cer	25000	background	ECF_2	12036	0.000269227	3.8676e-05	0.0130037	0.00726584	0.0547152	0.0547021	-0.000238775	15	15
cer	25000	background	ECF_3	12036	-0.00117671	-0.000417262	0.0558331	0.0357111	0.797065	0.797227	0.000203402	2	2
cer	25000	background	C_2	12036	0.000198196	-2.9739e-05	0.0121877	0.00730854	0.114071	0.114086	0.000134158	14	14
cer	25000	background	D_2	12036	0.000559775	0.00052407	0.0284603	0.0164205	0.36681	0.367244	0.00118353	6	6
cer	25000	background	L_2	12036	6.58822e-05	-3.72976e-05	0.0352316	0.023354	0.422166	0.422292	0.000297129	3	3
cer	25000	background	L_3	12036	0.00021616	0.000159446	0.0260056	0.0160744	0.198283	0.198648	0.00184128	12	12
cer	25000	background	Q_w	12036	0.000129118	0.000244159	0.0306836	0.0159664	0.252686	0.252954	0.00106207	10	10
cer	25000	background	ThrustMaj	12036	0.000322844	-5.25489e-05	0.0366636	0.0241381	0.372237	0.371912	-0.000873324	5	5
cer	50000	all	τ_1^{WTA}	50000	0.000108596	1.51917e-05	0.0197263	0.0122342	0.239834	0.239777	-0.000238583	11	11
cer	50000	all	τ_2^{WTA}	50000	0.000129714	3.27826e-07	0.0239885	0.0162414	0.290124	0.289725	-0.00137484	6	6
cer	50000	all	τ_3^{WTA}	50000	-5.4377e-05	1.78814e-05	0.0253191	0.0153071	0.267773	0.267859	0.00032265	9	9
cer	50000	all	τ_4^{WTA}	50000	0.00140543	0.00119638	0.0590088	0.040993	0.914582	0.914593	1.23826e-05	1	1
cer	50000	all	$\sqrt{d_{12}}$	50000	-0.000574548	-0.000192262	0.0318575	0.019349	0.263899	0.263739	-0.0006061	10	10
cer	50000	all	$\sqrt{d_{23}}$	50000	-0.000600141	-0.00018283	0.0381096	0.023311	0.381286	0.381133	-0.00040035	5	5
cer	50000	all	ECF_1	50000	0.000839648	0.000359278	0.027422	0.0136793	0.153806	0.154265	0.00297914	12	12
cer	50000	all	ECF_2	50000	0.00018847	-1.51899e-05	0.011001	0.00617491	0.0442916	0.0443595	0.00153321	15	15
cer	50000	all	ECF_3	50000	-0.00416904	-0.000940114	0.0618743	0.0372871	0.707757	0.70515	-0.00368353	2	2
cer	50000	all	C_2	50000	0.00018029	4.37079e-05	0.0107639	0.00646977	0.0965523	0.0964613	-0.000943126	14	14
cer	50000	all	D_2	50000	0.000184671	0.00014494	0.023984	0.013858	0.285472	0.285616	0.0005034	7	7
cer	50000	all	L_2	50000	-0.000995972	-0.000728875	0.0368574	0.0249911	0.413064	0.412367	-0.00168714	3	3
cer	50000	all	L_3	50000	-1.79546e-05	-7.04452e-06	0.019963	0.0117931	0.136902	0.137075	0.00126293	13	13
cer	50000	all	Q_w	50000	-0.000926045	-0.000162305	0.0327779	0.0178399	0.271318	0.270566	-0.002771	8	8
cer	50000	all	ThrustMaj	50000	-0.000765246	-0.000474215	0.0393922	0.0264041	0.410588	0.409488	-0.00267815	4	4
cer	50000	signal	τ_1^{WTA}	25930	-5.20128e-05	3.37586e-05	0.0184482	0.0113709	0.198912	0.199026	0.000569715	10	10
cer	50000	signal	τ_2^{WTA}	25930	-0.000393924	-0.000167428	0.0235412	0.0163076	0.25919	0.258816	-0.00144418	8	8
cer	50000	signal	τ_3^{WTA}	25930	-0.00048435	-0.000236131	0.026971	0.0165165	0.297318	0.297453	0.000451768	6	6
cer	50000	signal	τ_4^{WTA}	25930	0.00103681	0.000790656	0.0576229	0.0406968	0.87653	0.877213	0.000779832	1	1
cer	50000	signal	$\sqrt{d_{12}}$	25930	-0.00152333	-0.000690246	0.028119	0.0171156	0.197736	0.197118	-0.00312588	11	11
cer	50000	signal	$\sqrt{d_{23}}$	25930	-0.00122334	-0.000730351	0.0358515	0.0229286	0.351954	0.350859	-0.00311153	5	5
cer	50000	signal	ECF_1	25930	-9.74473e-05	0.000200095	0.0231713	0.0128123	0.14511	0.145816	0.00486694	12	12
cer	50000	signal	ECF_2	25930	8.79247e-05	-4.09938e-05	0.00893207	0.00529561	0.0342419	0.0344034	0.00471826	15	15
cer	50000	signal	ECF_3	25930	-0.00758767	-0.00191398	0.0674587	0.0387583	0.625941	0.621249	-0.0074967	2	2
cer	50000	signal	C_2	25930	1.66754e-05	3.02698e-05	0.00944977	0.00586437	0.0799064	0.079832	-0.000931761	14	14
cer	50000	signal	D_2	25930	-0.00012759	6.80238e-06	0.0194259	0.011914	0.206775	0.20701	0.0011374	9	9
cer	50000	signal	L_2	25930	-0.00229386	-0.0014125	0.0381293	0.0263509	0.407999	0.406588	-0.00345621	4	4
cer	50000	signal	L_3	25930	-0.000417158	-8.38134e-05	0.0146257	0.00899171	0.0819351	0.0823149	0.00463529	13	13
cer	50000	signal	Q_w	25930	-0.00184646	-0.000570099	0.0352425	0.0200367	0.289738	0.288094	-0.00567517	7	7
cer	50000	signal	ThrustMaj	25930	-0.00202524	-0.00113289	0.041985	0.0288711	0.446876	0.444806	-0.00463165	3	3
cer	50000	background	τ_1^{WTA}	24070	0.000281616	-2.30968e-06	0.0211032	0.0132055	0.283917	0.283679	-0.000837542	9	9

Systematic	Size	Subset	Feature	n	Mean ΔSHAP	Median ΔSHAP	Mean |ΔSHAP|	Median |ΔSHAP|	Nom. mean |SHAP|	Sys. mean |SHAP|	Rel. attr. drift	Nom. rank	Sys. rank
cer	50000	background	τ_2^{WTA}	24070	0.000693816	0.000202976	0.0244703	0.0161218	0.323442	0.32302	-0.00130536	8	8
cer	50000	background	τ_3^{WTA}	24070	0.000408821	0.000239246	0.0235396	0.0141202	0.235943	0.235979	0.000153027	11	11
cer	50000	background	τ_4^{WTA}	24070	0.00180252	0.0015825	0.0605017	0.0413277	0.955577	0.954876	-0.000732975	1	1
cer	50000	background	$\sqrt{d_{12}}$	24070	0.000447555	0.000366077	0.0358848	0.0223795	0.335174	0.335508	0.00099737	7	7
cer	50000	background	$\sqrt{d_{23}}$	24070	7.121e-05	0.000314899	0.0405422	0.0237489	0.412888	0.413744	0.00207193	4	4
cer	50000	background	ECF_1	24070	0.00184916	0.00052565	0.0320011	0.0146886	0.163176	0.163368	0.0011772	13	13
cer	50000	background	ECF_2	24070	0.000296784	1.94358e-05	0.0132299	0.0073375	0.0551183	0.0550847	-0.00060842	15	15
cer	50000	background	ECF_3	24070	-0.000486243	-7.4923e-05	0.0558585	0.0358	0.795892	0.795532	-0.000451739	2	2
cer	50000	background	C_2	24070	0.000356547	6.8143e-05	0.0121795	0.00724084	0.114485	0.114377	-0.000945991	14	14
cer	50000	background	D_2	24070	0.000521062	0.000299344	0.0288942	0.0167477	0.370246	0.370294	0.00013056	6	6
cer	50000	background	L_2	24070	0.000402207	-9.84436e-05	0.0354872	0.023568	0.418519	0.418592	0.000173679	3	3
cer	50000	background	L_3	24070	0.000412098	0.000147335	0.0257128	0.0160707	0.196116	0.196069	-0.000241013	12	12
cer	50000	background	Q_w	24070	6.54954e-05	0.000202347	0.0301229	0.0156504	0.251474	0.251683	0.000830166	10	10
cer	50000	background	ThrustMaj	24070	0.000592115	9.41716e-05	0.036599	0.0238961	0.371494	0.37144	-0.00014408	5	5
cer	100000	all	τ_1^{WTA}	100000	7.69385e-05	2.57641e-05	0.0197975	0.0122643	0.239835	0.239808	-0.000113761	11	11
cer	100000	all	τ_2^{WTA}	100000	4.49714e-05	-3.64333e-05	0.023961	0.0161316	0.289812	0.289637	-0.000604453	6	6
cer	100000	all	τ_3^{WTA}	100000	-2.48832e-05	7.33463e-05	0.0252972	0.0152809	0.266812	0.266816	1.5526e-05	9	9
cer	100000	all	τ_4^{WTA}	100000	0.00111636	0.00103918	0.0586034	0.0407436	0.913843	0.913847	3.91345e-06	1	1
cer	100000	all	$\sqrt{d_{12}}$	100000	-0.000610889	-0.000222564	0.0316628	0.0192531	0.264029	0.264112	0.000313906	10	10
cer	100000	all	$\sqrt{d_{23}}$	100000	-0.000671922	-0.000355348	0.0377278	0.0231882	0.381687	0.380972	-0.00187455	5	5
cer	100000	all	ECF_1	100000	0.00099967	0.000358012	0.0274771	0.0136107	0.153926	0.154481	0.00360355	12	12
cer	100000	all	ECF_2	100000	0.000171945	-5.82356e-06	0.0109505	0.00616603	0.0442861	0.0444161	0.00293499	15	15
cer	100000	all	ECF_3	100000	-0.00437206	-0.00108583	0.0617407	0.0371051	0.708505	0.70588	-0.00370589	2	2
cer	100000	all	C_2	100000	0.000136103	7.97752e-05	0.0107965	0.00650782	0.0960965	0.0959975	-0.00103118	14	14
cer	100000	all	D_2	100000	0.00021342	3.8527e-05	0.0240104	0.0138465	0.285772	0.2858	9.95943e-05	7	7
cer	100000	all	L_2	100000	-0.00133374	-0.000734501	0.0366764	0.024738	0.412908	0.412193	-0.00173	3	3
cer	100000	all	L_3	100000	-9.63784e-05	1.11759e-08	0.0199739	0.0117524	0.137674	0.137922	0.00179496	13	13
cer	100000	all	Q_w	100000	-0.00115803	-0.000327706	0.0325667	0.0177252	0.271597	0.270842	-0.00278023	8	8
cer	100000	all	ThrustMaj	100000	-0.000983684	-0.000457016	0.0391804	0.0263127	0.410739	0.409584	-0.00281183	4	4
cer	100000	signal	τ_1^{WTA}	51859	-0.000110159	3.55393e-05	0.0185117	0.0114388	0.199527	0.199586	0.000299701	10	10
cer	100000	signal	τ_2^{WTA}	51859	-0.000437295	-0.00024116	0.0233586	0.0162019	0.259082	0.259138	0.000215912	8	8
cer	100000	signal	τ_3^{WTA}	51859	-0.000360852	-0.000111759	0.0270019	0.0164175	0.296127	0.296084	-0.000147136	6	6
cer	100000	signal	τ_4^{WTA}	51859	0.000598283	0.000595808	0.0573848	0.0406569	0.87508	0.875504	0.000485103	1	1
cer	100000	signal	$\sqrt{d_{12}}$	51859	-0.00112158	-0.000502361	0.0279876	0.0169855	0.197473	0.197164	-0.00156532	11	11
cer	100000	signal	$\sqrt{d_{23}}$	51859	-0.00168402	-0.000931561	0.0357995	0.0229609	0.351449	0.350178	-0.00361741	5	5
cer	100000	signal	ECF_1	51859	0.00020109	0.000201166	0.0231424	0.0128012	0.145607	0.146358	0.0051559	12	12
cer	100000	signal	ECF_2	51859	0.000173929	-1.98353e-05	0.00886352	0.00530188	0.0342052	0.0344253	0.00643495	15	15
cer	100000	signal	ECF_3	51859	-0.00785092	-0.00208998	0.0672213	0.0384979	0.625352	0.620138	-0.00833747	2	2
cer	100000	signal	C_2	51859	5.31464e-08	5.84871e-05	0.00945272	0.0058731	0.0794096	0.0793443	-0.000822185	14	14
cer	100000	signal	D_2	51859	-0.000176278	-4.57261e-05	0.0194656	0.0119233	0.207659	0.20792	0.00125713	9	9
cer	100000	signal	L_2	51859	-0.00273015	-0.00152744	0.0380728	0.0261984	0.407673	0.406212	-0.0035839	4	4
cer	100000	signal	L_3	51859	-0.000434441	-9.18433e-05	0.0146859	0.00905939	0.082159	0.0825421	0.0046632	13	13
cer	100000	signal	Q_w	51859	-0.00219434	-0.000742391	0.0351444	0.0201252	0.28985	0.288348	-0.00518284	7	7
cer	100000	signal	ThrustMaj	51859	-0.00229703	-0.00102568	0.041846	0.0288827	0.447544	0.445534	-0.00448982	3	3
cer	100000	background	τ_1^{WTA}	48141	0.000278485	1.42306e-05	0.0211827	0.0132126	0.283259	0.283134	-0.000440419	9	9
cer	100000	background	τ_2^{WTA}	48141	0.000564484	0.00015983	0.0246099	0.01606	0.322914	0.322496	-0.00129458	8	8
cer	100000	background	τ_3^{WTA}	48141	0.000337033	0.000216484	0.0234609	0.0141137	0.235234	0.23529	0.000239322	11	11
cer	100000	background	τ_4^{WTA}	48141	0.00167446	0.0014921	0.0599162	0.0408385	0.955594	0.955135	-0.00047966	1	1
cer	100000	background	$\sqrt{d_{12}}$	48141	-6.07547e-05	0.000100762	0.035622	0.0221969	0.335721	0.33623	0.00151728	7	7

Systematic	Size	Subset	Feature	n	Mean ΔSHAP	Median ΔSHAP	Mean |ΔSHAP|	Median |ΔSHAP|	Nom. mean |SHAP|	Sys. mean |SHAP|	Rel. attr. drift	Nom. rank	Sys. rank
cer	100000	background	$\sqrt{d_{23}}$	48141	0.000418345	0.000218093	0.039805	0.0234137	0.41426	0.414145	-0.000277405	4	4
cer	100000	background	ECF_1	48141	0.00185993	0.000543959	0.0321465	0.014623	0.162889	0.163231	0.00210396	13	13
cer	100000	background	ECF_2	48141	0.000169808	1.28262e-05	0.0131986	0.00737221	0.0551456	0.0551782	0.000589811	15	15
cer	100000	background	ECF_3	48141	-0.000624525	-0.000148237	0.0558368	0.0357076	0.798075	0.798241	0.000207626	2	2
cer	100000	background	C_2	48141	0.000282661	0.000100821	0.0122441	0.0073145	0.114073	0.113935	-0.0012136	14	14
cer	100000	background	D_2	48141	0.000633215	0.000163287	0.0289063	0.0167862	0.369919	0.369694	-0.000609792	6	6
cer	100000	background	L_2	48141	0.000170528	-2.08318e-05	0.035172	0.0232022	0.418551	0.41864	0.000213255	3	3
cer	100000	background	L_3	48141	0.000267794	0.000159144	0.0256704	0.0160285	0.197478	0.197578	0.000503149	12	12
cer	100000	background	Q_w	48141	-4.16886e-05	5.04553e-05	0.0297899	0.0155084	0.251934	0.251992	0.000227361	10	10
cer	100000	background	ThrustMaj	48141	0.000431095	4.40031e-05	0.036309	0.0238565	0.371087	0.370861	-0.000608837	5	5

Table S7: Score-level and rank-level summaries of the paired SHAP-Delta campaign. Each row is one systematic, sample size and class subset. Score changes are in probability space. ρ is the Spearman correlation between the nominal and varied global feature-importance orderings within that subset; its minimum over all twelve systematic–subset combinations at 100,000 pairs is 0.986, and over the four all-jet combinations 0.993.

Systematic	Size	Subset	n	Mean Δscore	Median Δscore	Mean $\lvert\Delta\text{score}\rvert$	ρ
esup	25000	all	25000	0.00126594	0.000565549	0.0281417	0.996429
esup	25000	signal	12961	-0.00304732	-0.00116599	0.0321121	0.982143
esup	25000	background	12039	0.00590954	0.00141467	0.0238672	0.992857
esup	50000	all	50000	0.00131986	0.000569494	0.0281489	1
esup	50000	signal	25927	-0.00288043	-0.00108558	0.0320287	0.985714
esup	50000	background	24073	0.00584364	0.00139111	0.0239703	0.992857
esup	100000	all	100000	0.00116452	0.000552509	0.0279912	1
esup	100000	signal	51851	-0.00288173	-0.00102192	0.0319719	0.985714
esup	100000	background	48149	0.00552187	0.00131438	0.0237045	0.996429
esdown	25000	all	25000	-0.00804751	-0.0021117	0.0286176	0.989286
esdown	25000	signal	12963	-0.0110571	-0.00437677	0.0333686	1
esdown	25000	background	12037	-0.00480642	-0.00106848	0.0235012	0.996429
esdown	50000	all	50000	-0.00805625	-0.0021484	0.0290564	0.989286
esdown	50000	signal	25927	-0.0113878	-0.00467843	0.0338766	1
esdown	50000	background	24073	-0.0044681	-0.00107664	0.0238649	0.996429
esdown	100000	all	100000	-0.00805153	-0.00214166	0.0291714	0.992857
esdown	100000	signal	51853	-0.0113177	-0.00456274	0.0341224	1
esdown	100000	background	48147	-0.004534	-0.00109002	0.0238394	0.996429
cpos	25000	all	25000	1.68394e-05	0	0.00420883	1
cpos	25000	signal	12963	-2.23581e-05	0	0.00416909	1
cpos	25000	background	12037	5.90523e-05	0	0.00425163	1
cpos	50000	all	50000	5.47692e-05	0	0.00420677	1
cpos	50000	signal	25926	-4.53028e-05	0	0.00409154	1
cpos	50000	background	24074	0.00016254	0	0.00433087	1
cpos	100000	all	100000	3.68065e-05	0	0.00416695	1
cpos	100000	signal	51852	-4.37585e-05	0	0.00407881	1
cpos	100000	background	48148	0.000123569	0	0.00426188	1
cer	25000	all	25000	-0.000331825	-6.64443e-05	0.0211876	1
cer	25000	signal	12964	-0.0011782	-0.000375271	0.0224901	0.996429
cer	25000	background	12036	0.000579811	6.04306e-05	0.0197848	1
cer	50000	all	50000	-0.000223641	-3.82895e-05	0.021367	1
cer	50000	signal	25930	-0.00128461	-0.000351995	0.0226897	1
cer	50000	background	24070	0.000919317	8.38981e-05	0.0199422	1
cer	100000	all	100000	-0.000464717	-6.22502e-05	0.0211959	1
cer	100000	signal	51859	-0.00154458	-0.000404835	0.0224571	1
cer	100000	background	48141	0.000698547	7.55626e-05	0.0198373	1

S6 Statistical precision of the binned uncertainties

Every jet of every evaluated sample was given an independent Poisson weight of unit mean, except that a jet of `esup`, `esdown`, `cer` or `cpos` with an exact nominal partner takes its partner's weight, and the ISR/FSR variations and the nominal complements of the modelling samples reuse the nominal weights. The full frozen chain was recomputed for each of 100 replicas (base seed 20261009). Table S8 gives the standard deviation of every entry of Tables S2 and S3 over the replicas. Because each component is a modulus, entries whose measured value is comparable to their standard deviation have a skewed replica distribution, and their replica mean lies above the measured value.

Table S8: Statistical standard deviation, over 100 Poisson bootstrap replicas, of every binned quantity of Tables S2 and S3. The nominal background rejection column is an absolute standard deviation; every other column is in percentage points of relative uncertainty. Exactly paired `esup`, `esdown`, `cer` and `cpos` jets share the weight of their nominal partner, the ISR/FSR variations and the nominal complements of the modelling samples reuse the nominal weights, and all other samples are resampled independently. Above 1.55 TeV the signal-modelling term is the extrapolated maximum, so its standard deviation is that of the bin in which the maximum is measured.

p_{T} low	p_{T} high	Nominal	BR	CES	CER	CPOS	Cluster	Track eff.	Fake rate	Bias	Track	Sig. model.	Bkg. model.	Modelling	Sig. ISR	Sig. FSR	Bkg. ISR	Bkg. FSR	ISR/FSR	Total
350	550	1.16	1.12	1.30	0.96	1.54	1.82	1.73	1.78	2.16	0.55	1.54	1.16	0.30	1.42	0.76	1.55	1.40	1.28	
550	750	0.93	1.55	1.69	1.05	1.73	2.07	1.98	1.63	2.48	1.04	1.23	1.11	0.28	3.28	1.29	3.27	3.14	2.37	
750	950	0.73	1.29	1.53	1.17	1.66	1.94	2.00	2.33	2.57	1.43	1.11	1.16	0.17	2.15	1.01	2.10	2.07	1.81	
950	1150	0.73	1.36	1.08	0.90	1.41	1.55	1.99	1.46	1.75	2.33	1.93	2.23	0.16	2.05	0.76	2.16	2.07	1.79	
1150	1350	0.71	1.40	1.53	1.09	1.70	1.60	1.85	1.91	1.64	4.65	1.19	4.33	0.17	2.12	1.26	2.26	2.08	2.47	
1350	1550	0.68	1.35	1.42	1.08	1.63	1.79	1.90	1.64	1.98	4.65	1.41	3.45	0.16	2.26	0.95	1.93	2.25	2.57	
1550	1750	0.75	1.42	1.50	0.96	1.72	1.61	1.68	1.38	1.83	0.55	1.85	0.77	0.15	1.50	1.20	2.37	1.78	1.42	
1750	1950	0.53	1.22	1.31	1.11	1.57	1.38	1.45	1.44	1.50	0.55	1.55	0.71	0.15	2.38	1.10	2.09	2.25	1.42	
1950	2150	0.56	1.42	1.25	1.10	1.49	1.38	1.93	1.70	1.92	0.55	1.75	0.71	0.18	2.97	1.05	1.56	2.83	1.82	
2150	2350	0.67	1.47	1.56	1.17	1.81	2.03	2.12	2.20	2.67	0.55	2.14	0.79	0.14	3.43	1.22	1.75	3.37	1.93	
2350	2550	0.52	1.30	1.26	0.89	1.49	1.52	2.20	1.42	2.43	0.55	2.83	0.98	0.14	2.69	0.93	2.05	2.69	1.50	
2550	2750	0.40	1.34	1.26	0.77	1.47	1.25	1.71	1.28	1.41	0.55	2.34	1.01	0.09	2.81	1.15	1.18	2.72	1.62	
2750	2950	0.51	1.33	1.57	1.13	1.87	1.68	2.07	1.95	2.00	0.55	2.36	0.99	0.08	2.87	1.24	1.45	2.73	1.54	
2950	3150	0.40	1.54	1.06	1.00	1.42	1.80	2.23	1.38	2.27	0.55	3.15	1.16	0.08	1.83	1.04	1.24	1.86	1.50	

S7 Training-seed and input-set runs

Table S9: Every training run of the seed and input-set study. All eighteen runs use the frozen configuration and the same 4,499,612 fitting jets; only the training seed and the input set change. The seed-42 run on all fifteen inputs is the frozen model: its SHA-256 and its predictions on every sample are identical to the frozen ones. Uncertainties are inclusive relative uncertainties on background rejection at 50% signal efficiency, in percent; CES is the cluster energy-scale envelope.

Inputs	Seed	AUC	BR@50%	BR@80%	CES	Cluster	Track	Modelling	ISR/FSR	Total
All fifteen (frozen inputs)	42	0.93199	41.91	9.288	9.58	10.07	4.51	20.01	12.04	25.83
	1	0.93198	41.74	9.283	9.35	9.84	4.38	20.64	12.19	26.28
	2	0.93193	41.78	9.278	9.59	10.03	4.23	21.21	12.69	27.01
	3	0.93198	41.75	9.296	9.33	9.84	4.33	20.52	11.71	25.96
	4	0.93200	41.86	9.286	9.41	9.86	4.19	21.04	12.61	26.76
	5	0.93199	41.88	9.284	9.42	9.84	4.31	20.49	12.06	26.09
Nine dimensionless ($d = 0$)	42	0.87608	18.88	4.589	2.03	2.78	2.66	6.38	14.03	15.89
	1	0.87601	18.86	4.584	1.77	2.43	2.79	6.39	13.75	15.61
	2	0.87605	18.90	4.595	2.20	2.94	3.02	6.71	13.82	15.93
	3	0.87607	18.87	4.592	2.05	2.86	2.81	6.94	13.82	15.98
	4	0.87606	18.90	4.593	2.04	2.69	2.73	6.72	13.89	15.90
	5	0.87608	18.88	4.592	2.08	2.80	2.73	6.48	13.54	15.51
Nine + five X/ECF_1^d	42	0.87795	19.46	4.687	2.11	2.86	2.44	6.65	14.30	16.22
	1	0.87792	19.51	4.685	2.12	2.84	2.68	6.31	14.17	16.00
	2	0.87792	19.46	4.686	2.25	3.01	2.44	6.48	14.12	16.01
	3	0.87795	19.44	4.685	2.26	3.00	2.47	6.39	14.39	16.22
	4	0.87791	19.48	4.683	2.11	2.78	2.57	6.57	14.47	16.34
	5	0.87793	19.52	4.679	2.10	2.88	2.31	6.32	14.20	15.97

S8 Per-jet response of the inputs

For every exactly paired jet the relative response $r = X^s/X^{\mathrm{nom}} - 1$ of each reconstructed input was computed on all matched eligible pairs. The degree d is the power of a common constituent four-momentum rescaling under which the quantity scales.

Table S10: Per-jet relative response $X^s/X^{\mathrm{nom}} - 1$ of every input on every exactly paired calorimeter variation, in percent: median, mean, 16th and 84th percentiles, median absolute value, and the percentage of jets for which the quantity is unchanged. "Excl." counts pairs in which the quantity is non-finite on either side or zero on the nominal side and is therefore left out.

Sample	Quantity	d	Pairs	Excl.	Median	Mean	16%	84%	Median $\lvert r\rvert$	Unchanged
`esup`	τ_1^{WTA}	0	4,999,999	0	-0.939	-1.405	-3.307	+0.023	1.129	0.0004
	τ_2^{WTA}	0	4,999,999	0	-1.447	-1.746	-3.605	-0.125	1.524	0.0003
	τ_3^{WTA}	0	4,999,999	0	-1.727	-1.919	-3.663	-0.282	1.772	0.0001
	τ_4^{WTA}	0	4,999,956	43	-1.874	-2.020	-3.760	-0.352	1.916	0.0001
	$\sqrt{d_{12}}$	1	4,999,999	0	+5.417	+5.506	+1.863	+9.000	5.493	0.0000
	$\sqrt{d_{23}}$	1	4,999,999	0	+4.287	+5.207	+1.758	+8.855	4.533	0.0041
	ECF_1	1	4,999,999	0	+7.407	+6.781	+3.724	+9.301	7.407	0.0000
	ECF_2	2	4,999,999	0	+12.571	+12.686	+7.023	+18.629	12.571	0.0000
	ECF_3	3	4,999,999	0	+17.629	+18.438	+10.062	+28.059	17.629	0.0000
	C_2	0	4,999,999	0	-0.324	-0.441	-1.038	+0.144	0.422	0.0005
	D_2	0	4,999,999	0	+0.284	+0.796	-0.450	+2.492	0.696	0.0007
	L_2	0	4,999,999	0	-0.417	-0.643	-1.528	+0.103	0.521	0.0008
	L_3	0	4,999,999	0	-1.000	-1.461	-2.933	-0.141	1.025	0.0004
	Q_w	1	4,999,999	0	+5.484	+6.066	+2.138	+8.891	5.561	0.0030
	Thrust major	0	4,999,999	0	+0.077	+0.100	-0.130	+0.373	0.154	0.0014
`esdown`	τ_1^{WTA}	0	4,999,999	0	+1.080	+1.643	-0.037	+3.797	1.294	0.0006
	τ_2^{WTA}	0	4,999,999	0	+1.650	+2.028	+0.124	+4.138	1.733	0.0003
	τ_3^{WTA}	0	4,999,999	0	+1.978	+2.228	+0.300	+4.210	2.024	0.0001
	τ_4^{WTA}	0	4,999,956	43	+2.145	+2.341	+0.375	+4.315	2.187	0.0002
	$\sqrt{d_{12}}$	1	4,999,999	0	-5.406	-5.315	-9.005	-1.856	5.489	0.0000
	$\sqrt{d_{23}}$	1	4,999,999	0	-4.297	-4.914	-8.868	-1.751	4.568	0.0042
	ECF_1	1	4,999,999	0	-7.407	-6.781	-9.301	-3.724	7.407	0.0000
	ECF_2	2	4,999,999	0	-11.922	-11.871	-17.045	-6.822	11.922	0.0000
	ECF_3	3	4,999,999	0	-15.938	-16.230	-23.648	-9.506	15.938	0.0000
	C_2	0	4,999,999	0	+0.365	+0.499	-0.155	+1.159	0.470	0.0007
	D_2	0	4,999,999	0	-0.324	-0.833	-2.654	+0.502	0.778	0.0007
	L_2	0	4,999,999	0	+0.469	+0.728	-0.116	+1.714	0.581	0.0006
	L_3	0	4,999,999	0	+1.132	+1.672	+0.156	+3.319	1.157	0.0001
	Q_w	1	4,999,999	0	-5.555	-5.437	-8.902	-2.138	5.638	0.0033
	Thrust major	0	4,999,999	0	-0.086	-0.109	-0.415	+0.150	0.175	0.0011
`cer`	τ_1^{WTA}	0	4,995,058	0	-0.062	+0.073	-3.385	+3.394	2.125	0.0001
	τ_2^{WTA}	0	4,995,058	0	-0.095	+0.044	-3.747	+3.715	2.361	0.0000
	τ_3^{WTA}	0	4,995,058	0	-0.119	+0.019	-3.808	+3.744	2.407	0.0001
	τ_4^{WTA}	0	4,995,015	43	-0.162	-0.035	-3.927	+3.768	2.464	0.0001
	$\sqrt{d_{12}}$	1	4,995,058	0	+0.007	+0.258	-4.174	+4.202	2.674	0.0001
	$\sqrt{d_{23}}$	1	4,995,058	0	-0.012	+0.440	-5.093	+5.105	3.202	0.0040
	ECF_1	1	4,995,058	0	0.000	-0.001	-3.293	+3.291	2.143	0.0001
	ECF_2	2	4,995,058	0	-0.063	-0.002	-5.460	+5.438	3.594	0.0001
	ECF_3	3	4,995,058	0	-0.158	-0.003	-7.389	+7.345	4.883	0.0001
	C_2	0	4,995,058	0	-0.023	-0.017	-1.194	+1.141	0.718	0.0005
	D_2	0	4,995,058	0	-0.005	+0.056	-2.278	+2.340	1.378	0.0002
	L_2	0	4,995,058	0	-0.044	-0.022	-1.595	+1.511	0.941	0.0003
	L_3	0	4,995,058	0	-0.071	-0.002	-2.411	+2.282	1.396	0.0001
	Q_w	1	4,995,058	0	-0.047	+0.803	-3.600	+3.477	2.266	0.0031
	Thrust major	0	4,995,058	0	+0.009	+0.014	-0.552	+0.586	0.326	0.0006
`cpos`	τ_1^{WTA}	0	4,999,999	0	0.000	+0.031	-0.057	+0.065	0.023	20.5988
	τ_2^{WTA}	0	4,999,999	0	0.000	+0.067	-0.129	+0.153	0.055	20.5923
	τ_3^{WTA}	0	4,999,999	0	0.000	+0.077	-0.206	+0.245	0.090	20.5910
	τ_4^{WTA}	0	4,999,955	44	0.000	+0.067	-0.260	+0.303	0.115	20.5907
	$\sqrt{d_{12}}$	1	4,999,999	0	0.000	+0.077	-0.064	+0.068	0.023	20.6010

Sample	Quantity	d	Pairs	Excl.	Median	Mean	16%	84%	Median $\lvert r \rvert$	Unchanged
	$\sqrt{d_{23}}$	1	4,999,999	0	0.000	+0.222	-0.127	+0.138	0.035	28.6351
	ECF_1	1	4,999,999	0	0.000	0.000	-0.003	+0.003	0.001	21.1793
	ECF_2	2	4,999,999	0	0.000	+0.020	-0.063	+0.073	0.027	20.5967
	ECF_3	3	4,999,999	0	0.000	+0.073	-0.253	+0.301	0.117	20.5894
	C_2	0	4,999,999	0	0.000	+0.027	-0.134	+0.160	0.064	20.5906
	D_2	0	4,999,999	0	0.000	+0.010	-0.106	+0.121	0.049	20.5920
	L_2	0	4,999,999	0	0.000	+0.041	-0.095	+0.111	0.042	20.5935
	L_3	0	4,999,999	0	0.000	+0.044	-0.086	+0.091	0.035	20.5948
	Q_w	1	4,999,999	0	0.000	+0.326	-0.114	+0.129	0.041	27.9834
	Thrust major	0	4,999,999	0	0.000	-0.003	-0.036	+0.033	0.013	20.6037